

Learning Quadrupedal Locomotion over Challenging Terrain

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Some of the most challenging environments on our planet are accessible to quadrupedal animals but remain out of reach for autonomous machines. Legged locomotion can dramatically expand the operational domains of robotics. However, conventional controllers for legged locomotion are based on elaborate state machines that explicitly trigger the execution of motion primitives and reflexes. These designs have escalated in complexity while falling short of the generality and robustness of animal locomotion. Here we present a radically robust controller for legged locomotion in challenging natural environments. We present a novel solution to incorporating proprioceptive feedback in locomotion control and demonstrate remarkable zero-shot generalization from simulation to natural environments. The controller is trained by reinforcement learning in simulation. It is based on a neural network that acts on a stream of proprioceptive signals. The trained controller has taken two generations of quadrupedal ANYmal robots to a variety of natural environments that are beyond the reach of prior published work in legged locomotion. The controller retains its robustness under conditions that have never been encountered during training: deformable terrain such as mud and snow, dynamic footholds such as rubble, and overground impediments such as thick vegetation and gushing water. The presented work opens new frontiers for robotics and indicates that radical robustness in natural environments can be achieved by training in much simpler domains.

1. INTRODUCTION

Legged locomotion can dramatically expand the reach of robotics. Much of the dry landmass on Earth remains impassible to wheeled and tracked machines, the stability of which can be severely compromised on challenging terrain. Quadrupedal animals, on the other hand, can access some of the most remote parts of our planet. They can choose safe footholds within their kinematic reach and rapidly change their kinematic state in response to the environment. Legged robots have the potential to traverse any terrain that their animal counterparts can.

To date, no published work has demonstrated dynamic locomotion in diverse, challenging natural environments as shown in Fig. 1. These environments have highly irregular profiles, deformable terrain, slippery surfaces, and overground obstructions. Under such conditions, existing published controllers manifest frequent foot slippage, loss of balance, and ultimately catastrophic failure. The challenge is exacerbated by the inaccessibility of veridical information about the physical properties of the terrain. Exteroceptive sensors such as cameras and LiDAR cannot reliably measure physical characteristics such as friction and compliance, are impeded by obstructions such as vegetation, snow, and water, and may not have the coverage and temporal resolution to capture changes induced by the robot itself, such as the crumbling of loose ground under the robot’s feet. Under these conditions, the robot must rely crucially on proprioception – the sensing of its own bodily

configuration at high temporal resolution. In response to unforeseen events such as unexpected ground contact, terrain deformation, and foot slippage, the controller must rapidly produce whole-body trajectories subject to multiple objectives: balancing, avoiding self-collision, counteracting external disturbances, and locomotion. While animals instinctively solve this complex control problem, it is an open challenge in robotics.

Conventional approaches to legged locomotion on uneven terrain have yielded increasingly complex control architectures. Many rely on elaborate state machines that coordinate the execution of motion primitives and reflex controllers [1–5]. To trigger transitions between states or the execution of a reflex, many systems explicitly estimate states such as ground contact and slippage [6–8]. Such estimation is commonly based on empirically tuned thresholds and can become erratic in the presence of unmodeled factors such as mud, snow, or vegetation. Other systems employ contact sensors at the feet, which can become unreliable in field conditions [9–11]. Overall, conventional systems for legged locomotion on rough terrain escalate in complexity as more scenarios are taken into account, have become extremely laborious to develop and maintain, and remain vulnerable to corner cases.

Model-free reinforcement learning (RL) has recently emerged as an alternative approach in the development of legged locomotion skills [12–14]. The idea of RL is to tune a controller to optimize a

在挑战性地形上学习四足运动

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地球上一些最具挑战性的环境, 四足动物可以进入, 但自主机器却难以企及。足式运动能够极大地拓展机器人学的操作领域。然而, 传统的足式运动控制器基于精心设计的状态机, 这些状态机显式地触发运动基元和反射的执行。这些设计在复杂性不断攀升的同时, 却未能达到动物运动的通用性和鲁棒性。在此, 我们提出了一种用于在具有挑战性的自然环境中实现足式运动的极端鲁棒控制器。我们提出了一种将本体感觉反馈融入运动控制的新颖解决方案, 并展示了从仿真到自然环境的卓越零样本泛化能力。该控制器通过仿真中的强化学习进行训练, 其基础是一个作用于本体感觉信号流的神经网络。经过训练的控制​​器已带领两代四足 ANYmal机器人进入了多种自然环境, 这些环境超出了此前已发表的足式运动研究工作的能力范围。该控制器在训练期间从未遇到过的条件下仍能保持其鲁棒性: 例如泥地和雪地等可变形地形、碎石等动态立足点, 以及茂密植被和湍急水流等地​​面障碍物。本项工作为机器人学开辟了新前沿, 并表明在更简单的领域中进行训练, 即可在自然环境中实现极端鲁棒性。

1. 引言

足式运动能够极大地拓展机器人学的应用范围。地球上大部分干燥陆地是轮式和履带式机器无法通行的, 这些机器在挑战性地形上的稳定性会受到严重损害。而四足动物则能够到达我们星球上一些最偏远的地区。它们可以在运动学可达范围内选择安全的立足点, 并迅速根据环境改变运动学状态。腿式机器人有潜力穿越其动物同类所能通行的任何地形。

迄今为止, 尚无已发表的研究成果展示过如图1所示, 在多样且具有挑战性的自然环境中实现动态运动。这些环境具有高度不规则的轮廓、可变形地形、湿滑表面以及地面障碍物。在此类条件下, 现有已发表的控制​​器会频繁出现足部打滑、失去平衡, 并最终导致灾难性故障。由于无法获取关于地形物理特性的真实信息, 这一挑战变得更加严峻。相机和激光雷达等外部传感器无法可靠地测量摩擦和柔顺性等物理特性, 会受到植被、雪​​地和积水等障碍物的阻碍, 并且可能无法覆盖或具备足够的时间分辨率来捕捉机器人自身引起的变化, 例如机器人足部下松软地面的崩塌。在这些条件下, 机器人必须主要依赖本体感觉——即对自身身体构型的感知。

以高时间分辨率进行感知。为了应对诸如意外地面接触、地形变形和足部打滑等不可预见事件, 控制器必须快速生成满足多个目标的全身轨迹: 平衡、避免自碰撞、抵抗外部干扰以及运动。虽然动物能本能地解决这一复杂的控制问题, 但在机器人学中, 这仍是一个悬而未决的挑战。

在不平坦地形上实现足式运动的传统方法催生了日益复杂的控制架构。许多方法依赖于精心设计的状态机来协调运动基元和反射控制器的执行 [1–5]。为了触发状态之间的转换或反射的执行, 许多系统会显式估计诸如地面接触与打滑等状态 [6–8]。这种估计通常基于经验调优的阈值, 在存在泥地、雪地或植被等未建模因素时可能变得不稳定。其他系统在足部使用接触传感器, 这些传感器在现场条件下可能变得不可靠 [9–11]。总体而言, 用于崎岖地形上足式运动的传统系统, 随着考虑的场景增多, 其复杂性不断升级, 开发和维护变得极其费力, 并且仍然容易受到边界情况的影响。

无模型强化学习 (RL) 最近已成为开发腿部运动技能的一种替代方法 [12–14]。强化学习的思想是调整控制器以优化

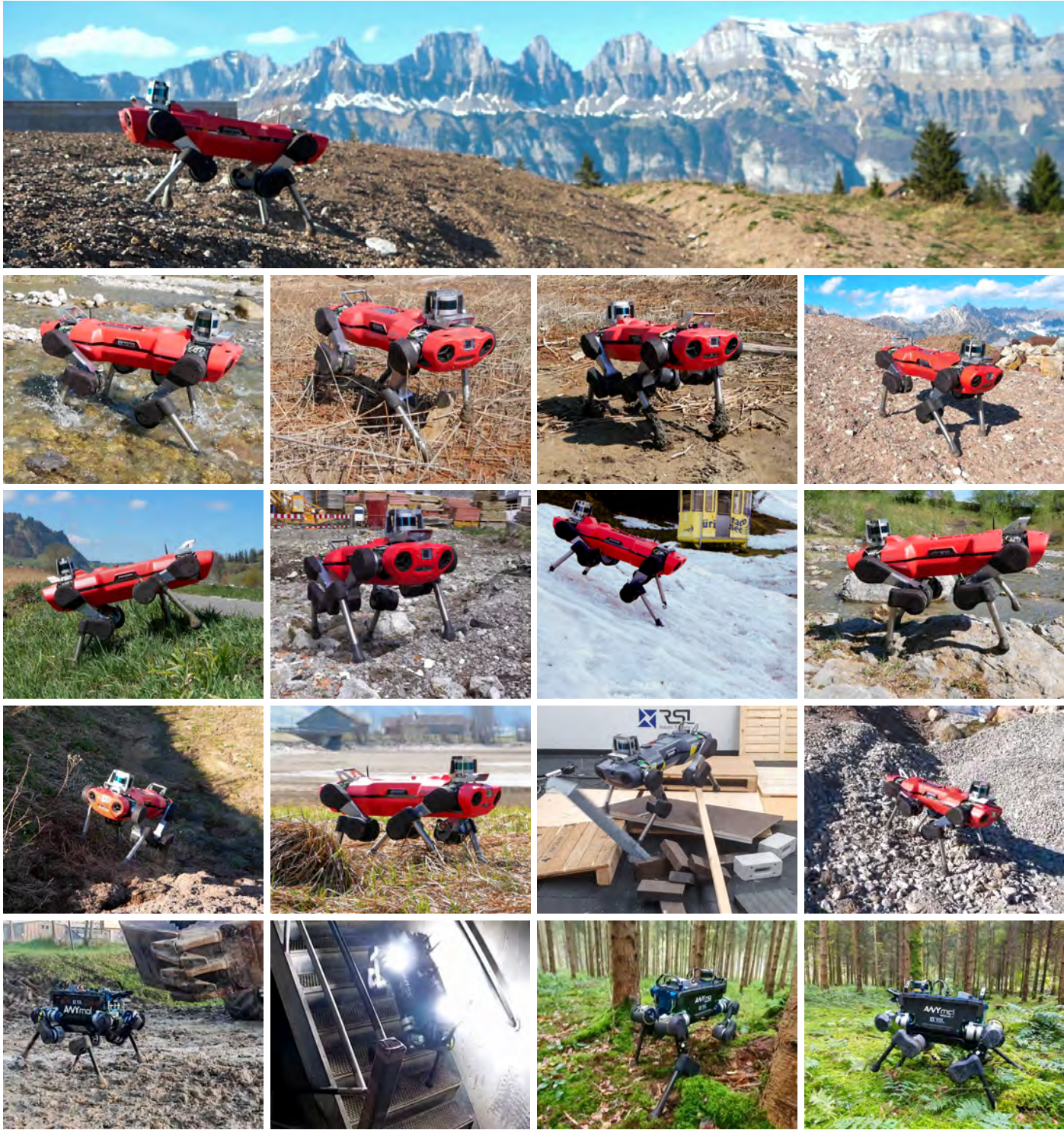


Fig. 1. Deployment of the presented locomotion controller in a variety of challenging environments.

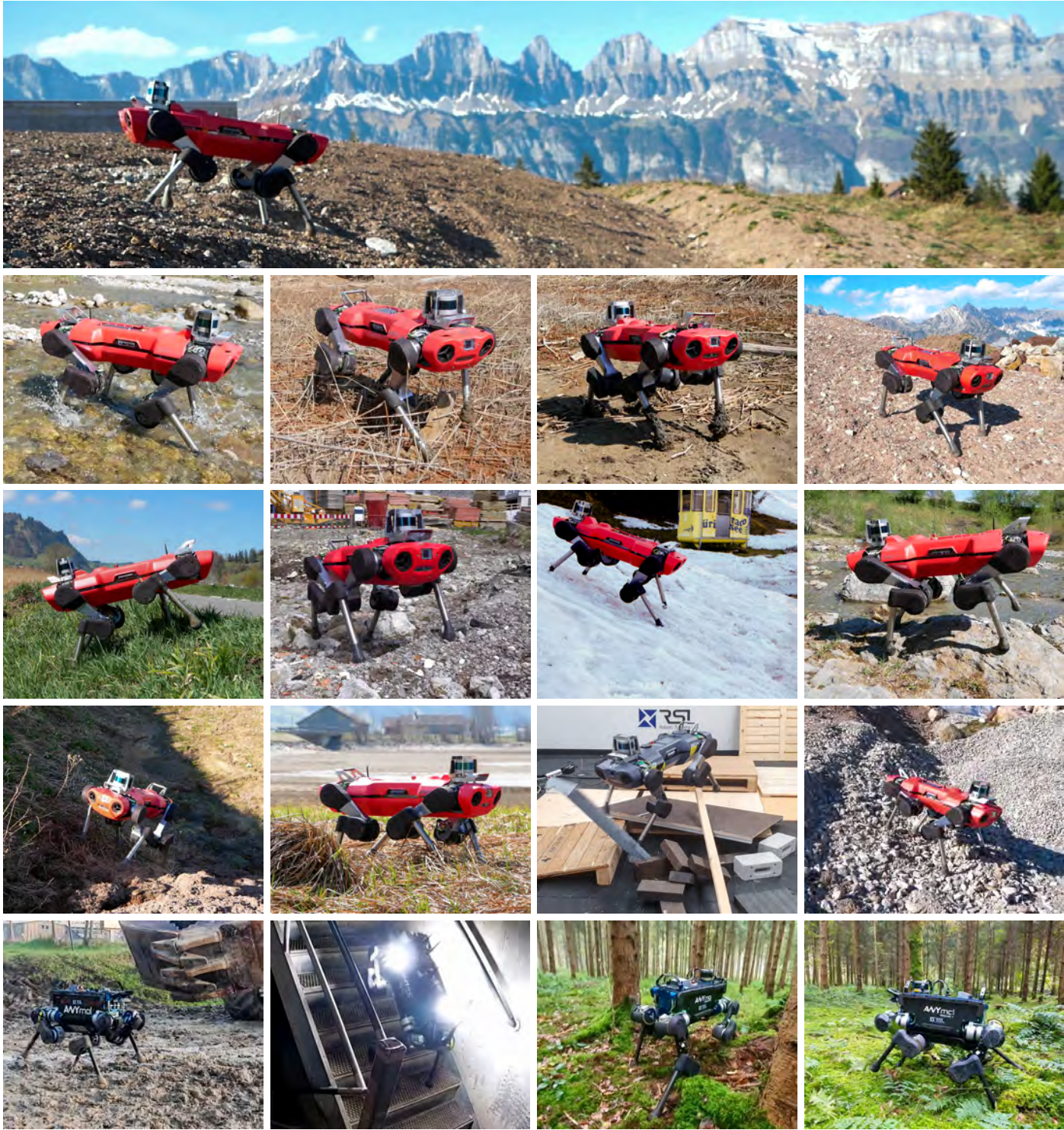


图 1. 所提出的运动控制器在多种具有挑战性环境中的部署

ts.

given reward function. The optimization is performed on data acquired by executing the controller itself, which improves with experience. RL has been used to simplify the design of locomotion controllers, automate parts of the design process, and learn behaviors that could not be engineered with prior approaches [12–15].

However, application of RL to legged locomotion has largely been confined to laboratory environments and conditions. Our prior work demonstrated end-to-end learning of locomotion and recovery behaviors – but only on flat ground, in the lab [12]. Other work also developed RL techniques for legged locomotion, but likewise focused largely on flat or moderately textured surfaces in laboratory settings [13, 14, 16–19].

Here we present a radically robust controller for blind quadrupedal locomotion on challenging terrain. The controller uses only proprioceptive measurements from joint encoders and an inertial measurement unit (IMU), which are the most durable and reliable sensors on legged machines. The operation of the controller is shown in Fig. 1 and Movie 1. The controller was used to drive two generations of ANYmal quadrupeds [20] in a variety of conditions that are beyond the reach of prior published work in legged robotics. The controller reliably trots through mud, sand, rubble, thick vegetation, snow, running water, and a variety of other off-road terrain. The same controller was also used in our entry in the DARPA Subterranean Challenge Urban Circuit. In all deployments, robots of the same generation were driven by exactly the same controller under all conditions. No tuning was required to adapt to different environments.

Like a number of prior applications of model-free RL to legged locomotion, we train the controller in simulation [12, 14, 16]. Prior efforts have established a number of practices for successful transfer of legged locomotion controllers from simulation to physical machines. One is realistic modeling of the physical system, including the actuators [12]. Another is randomization of physical parameters that vary between simulation and reality, such that the controller becomes robust to a range of conditions that cover those that arise in physical deployment, without the necessity to precisely model these conditions a priori [21].

We use these ideas as well, but have found that they were not sufficient to achieve robust locomotion on rough terrain. We therefore introduce and validate a number of additional ingredients that are crucial to realizing the presented skills. The first is a different policy architecture. Rather than using a multi-layer perceptron (MLP) that operates on a snapshot of the robot’s current state, as was common in prior work, we use a sequence model, specifically a temporal convolutional network (TCN) [22] that produces actuation based on an extended history of proprioceptive states. We do not employ explicit contact and slip estimation modules, which are known to be brittle in challenging situations; rather, the TCN learns to implicitly reason about contact and slippage events from proprioceptive history as needed.

The second important idea that enables the demonstrated results is privileged learning [23]. We have found that training a rough-terrain locomotion policy directly via reinforcement learning was not successful: the supervisory signal was sparse and the presented network failed to learn locomotion within reasonable time budgets. Instead, we decompose the training process into two stages. First, we train a teacher policy that has access to privileged information, namely ground-truth knowledge of the terrain and the robot’s contact with it. The privileged information enables the policy to quickly achieve high performance. We then use this privileged teacher to guide the learning of a purely proprioceptive student controller that only uses sensors that are available on the real robot. This privileged learning protocol is enabled by simulation, but the resulting proprioceptive policy is not confined to simulation and is deployed on physical machines.

The third idea that has proven important in achieving the presented



Fig. 2. A number of specific deployments. (A-F) Zero-shot generalization to slippery and deforming terrain. (G) Steep descent during the DARPA Subterranean Challenge. The stair rise is 18 cm and the slope is $\sim 45^\circ$.

levels of robustness is an automated curriculum that synthesizes terrains adaptively, based on the controller’s performance at different stages of the training process. In essence, terrains are synthesized such that the controller is capable of traversing them while becoming more robust. We evaluate the traversability of parameterized terrains and use particle filtering to maintain a distribution of terrain parameters of medium difficulty [24, 25] that adapt as the neural network learns. The training conditions grow increasingly more challenging, yielding an omnidirectional controller that combines agility with unprecedented resilience.

The result is a legged locomotion controller that is far more robust than any counterparts in existing literature. Remarkably, the controller is consistently effective in *zero-shot generalization* settings. That is, it remains robust when tested in conditions that have never been encountered during training. Our training in simulation only uses rigid terrains and a small set of procedurally generated terrain profiles, such as hills and steps. Yet when deployed on physical quadrupeds, the controller successfully handles deformable terrain (mud, moss, snow), dynamic footholds (stepping on a rolling board in a cluttered indoor environment, or debris in the field), and overground impediments such as thick vegetation, rubble, and gushing water. Our methodology and results open new frontiers for legged robotics and suggest that the extraordinary complexity of the physical world can be tamed without brittle and painstaking modeling or dangerous and expensive trial-and-error in field conditions.

2. RESULTS

Movie 1 summarizes the results of the presented work. We have deployed the trained locomotion controller on two generations of ANYmal robots: ANYmal-B (Fig. 2D-G) and ANYmal-C (Fig. 2A-C and Fig. 3). The robots have different kinematics, inertia, and actuators.

Natural environments

The presented controller has been deployed in diverse natural environments, as shown in Fig. 1 and Movies 1 and S1. These include steep

给定奖励函数。优化过程基于执行控制器自身所获取的数据进行，该控制器会随着经验积累而不断改进。强化学习已被用于简化运动控制器的设计、自动化设计流程中的部分环节，并学习那些无法通过先前方法工程化实现的行为 [12–15]。

然而，强化学习在足式运动中的应用在很大程度上仍局限于实验室环境和条件。我们先前的工作展示了运动与恢复行为的端到端学习——但仅限于实验室内的平坦地面 [12]。其他研究也开发了用于足式运动的强化学习技术，但同样主要聚焦于实验室环境中的平坦或中等纹理表面 [13, 14, 16–19]。

本文提出了一种用于挑战性地形上盲态四足运动的极其鲁棒的控制器。该控制器仅使用来自关节编码器和惯性测量单元的 proprioceptive 测量值，这些是腿式机器上最耐用、最可靠的传感器。控制器的运行情况如图1和视频1所示。该控制器被用于驱动两代ANYmal四足机器人 [20] 在多种超越先前腿式机器人学已发表工作范畴的条件下运行。该控制器能够可靠地在泥地、沙地、碎石、茂密植被、雪地、流水以及各种其他越野地形中小跑。同一控制器还被用于我们参加DARPA地下挑战赛城市赛段的机器人。在所有部署中，同代机器人在所有条件下均由完全相同的控制器驱动，无需针对不同环境进行任何调参。

与许多先前将无模型强化学习应用于足式运动的方法类似，我们在仿真中训练控制器 [12, 14, 16]。先前的工作已确立了一系列成功将腿式运动控制器从仿真迁移到物理机器的实践。其一是对物理系统（包括执行器 [12]）进行真实建模。其二是对仿真与现实之间变化的物理参数进行随机化，使得控制器能够对涵盖物理部署中出现的各种条件具有鲁棒性，而无需事先精确建模这些条件 [21]。

我们也采用了这些思路，但发现它们并不足以在崎岖地形上实现鲁棒运动。因此，我们引入并验证了若干对实现所展示技能至关重要的额外要素。首先是不同的策略架构。与先前工作中常见的基于机器人当前状态快照运行的多层感知机（MLP）不同，我们采用序列模型，具体来说时序卷积网络（TCN） [22]，它基于本体感知状态的扩展历史生成驱动指令。我们不使用显式的接触与滑移估计模块（这些模块在挑战性场景中往往表现脆弱）；相反，TCN会根据需要从本体感受历史中隐式学习推理接触和滑移事件。

第二个实现所展示结果的重要理念是特权学习 [23]。我们发现，直接通过强化学习训练崎岖地形运动策略并不成功：监督信号稀疏，且所提出的网络无法在合理的时间预算内学会运动。因此，我们将训练过程分解为两个阶段。首先，我们训练一个能够访问特权信息的教师策略，即地形的真实知识以及机器人与地形的接触情况。特权信息使策略能够快速达到高性能。然后，我们利用这个特权教师来指导一个纯本体感知学生控制器的学习，该控制器仅使用真实机器人上可用的传感器。这种特权学习协议通过仿真实现，但最终得到的本体感受策略并不局限于仿真，而是部署在物理机器上。

在实现所展示成果中，第三个被证明重要的理念是



图2. 若干特定部署场景。 (A-F) 对湿滑及可变形地形的零样本泛化。 (G) DARPA地下挑战赛城市赛段（下楼梯）。台阶高度为18厘米，斜率为 $\sim 45^\circ$ 。

本质上，地形被合成出来，使得控制器能够穿越它们，同时变得更加鲁棒。我们评估参数化地形的可穿越性，并利用粒子滤波来维持一个中等难度地形参数的分布 [24, 25]，该分布会随着神经网络的学习而自适应调整。训练条件逐渐变得更加具有挑战性，最终产生一个兼具敏捷性与前所未有的韧性的全向控制器。

其结果是，该腿式运动控制器比现有文献中的任何同类产品都更加鲁棒。值得注意的是，该控制器在零样本泛化设置中始终有效。也就是说，当在训练期间从未遇到过的条件下进行测试时，它仍然保持鲁棒性。我们的仿真训练仅使用刚性地形和一小部分程序化生成地形轮廓（如丘陵和台阶）。然而，当部署在物理四足机器人上时，该控制器成功处理了可变形地形（泥地、苔藓、雪地）、动态立足点（在杂乱的室内环境中踩踏滚动的木板，或现场碎片）以及地面障碍物，如茂密植被、碎石和湍急水流。我们的方法和结果为腿式机器人学开辟了新的前沿，并表明物理世界的非凡复杂性可以被驯服，而无需脆弱且费力的建模，也无需危险且昂贵的现场条件试错。

2. 结果

视频1总结了所提出工作的成果。我们已将训练好的运动控制器部署在两代ANYmal机器人上：ANYmal-B (图2D-G) 和ANYmal-C (图2A-C及图3)。这些机器人具有不同的运动学、惯性和执行器。

自然环境

所提出的控制器已部署在多种自然环境中，如图1和视频1及S1所示。这些环境包括陡峭的

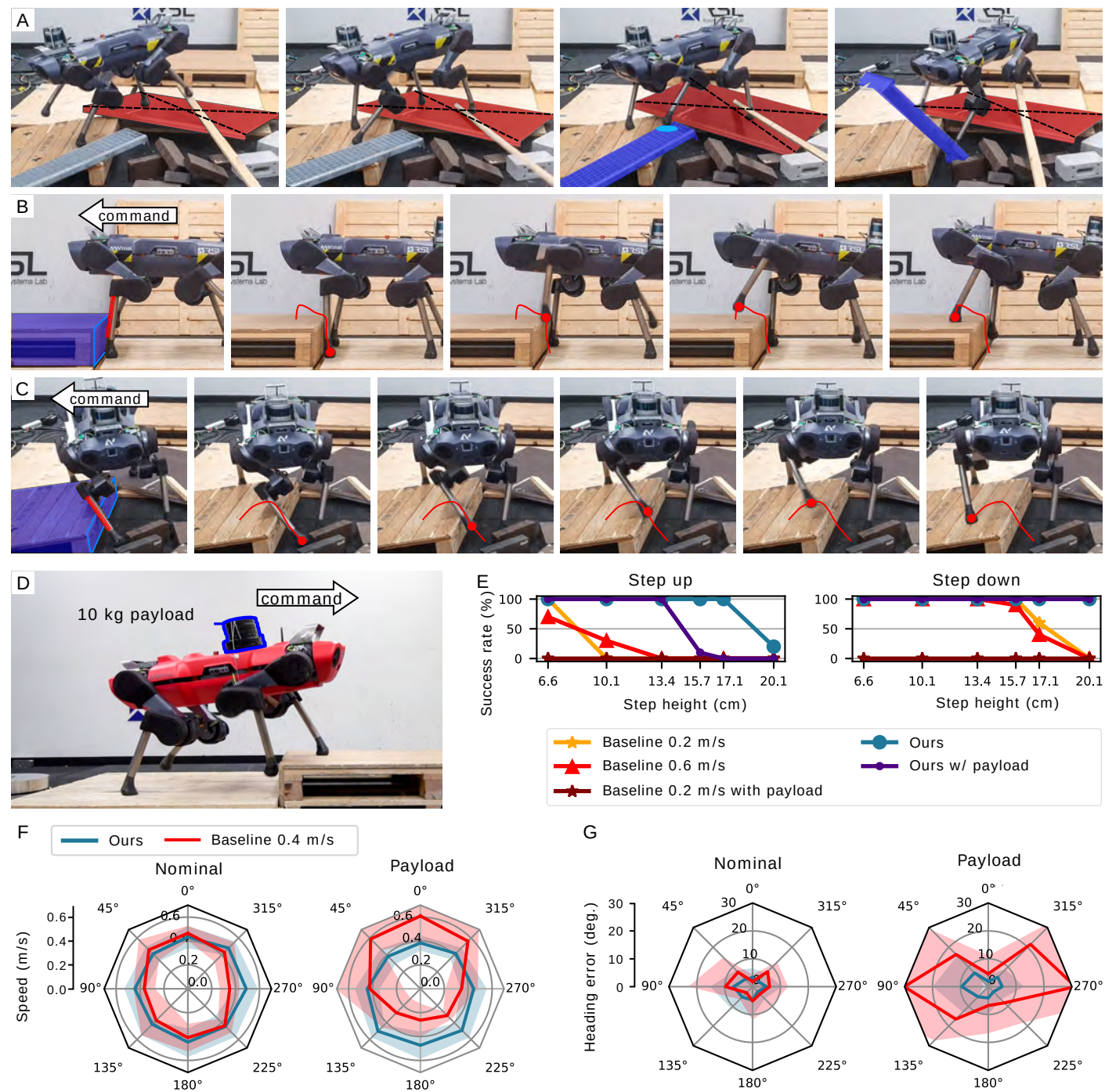


Fig. 3. Evaluation in an indoor environment. (A) Locomotion over unstable debris. The robot steps onto loose boards (highlighted in red and blue) that dislodge under the robot's feet. (B) The policy exhibits a foot-trapping reflex and overcomes a 16.8 cm step. (C) The policy learns to appropriately handle obstructions irrespective of the contact location. Here it is shown reacting to an obstacle that is encountered mid-shin during the swing phase. (D) Controlled experiments with steps and payload. Our controller and a baseline [1, 26] are commanded to walk over a step with and without the 10 kg payload. (E) Success rates for different step heights. The success rate is evaluated over 10 trials for each condition. (F) Mean linear speed for different command directions on flat terrain. 0° refers to the front of the robot. Shaded area denotes 95% confidence interval. (G) Mean heading errors for different command directions on flat terrain. Shaded area denotes 95% confidence interval.

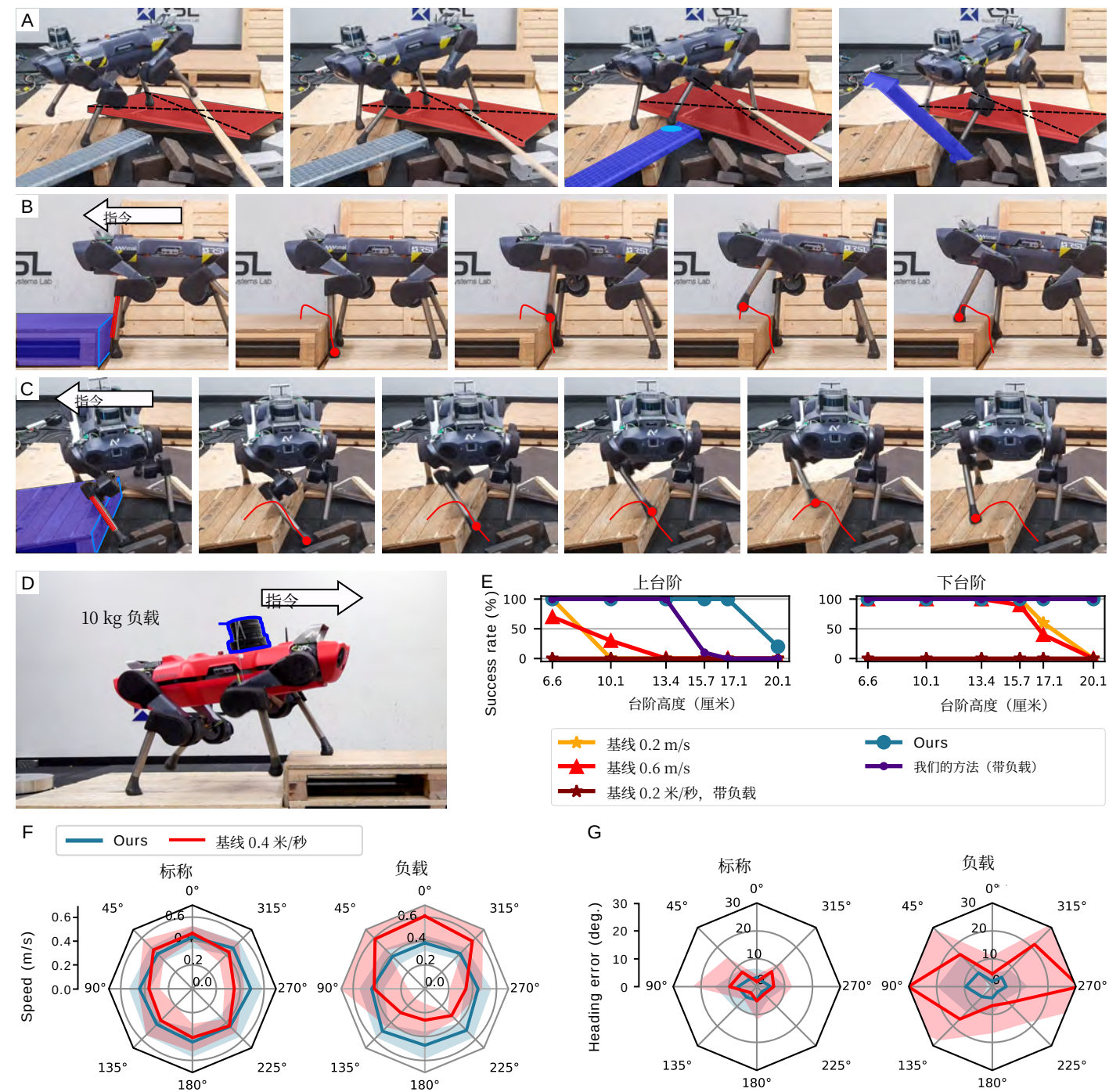


图 3. 室内环境评估。 (A) 不稳定碎片上的运动。机器人踏上松散木板（以红色和蓝色高亮显示），这些木板在机器人足部下松动脱落。(B) 该策略展现出足部卡住反射，并克服了 16.8 厘米高的台阶。(C) 该策略学会无论接触位置如何，都能适当处理障碍物。此处展示了它在摆动相期间小腿中部遇到障碍物时的反应。(D) 针对台阶和负载的控制实验。我们的控制器和基线 [1, 26] 被指令在有无 10 公斤负载的情况下越过台阶。(E) 不同台阶高度的成功率。每种条件下的成功率基于 10 次试验评估。(F) 平坦地形上不同指令方向的平均线速度。 0° 指机器人前方。阴影区域表示 95% 置信区间。(G) 平坦地形上不同指令方向的平均航向误差。阴影区域表示 95% 置信区间。

Quantity	Controller	Terrain		
		Moss	Mud	Vegetation
Average speed (m/s)	Ours	0.452	0.338	0.248
	Baseline	0.199	0.197	–
Average mechanical COT	Ours	0.423	0.692	1.23
	Baseline	0.625	0.931	–

Table 1. Comparison of locomotion performance in natural environments. The mechanical COT is computed using positive mechanical power exerted by the actuators.

mountain trails, creeks with running water, mud, thick vegetation, loose rubble, snow-covered hills, and a damp forest. A number of specific scenarios are further highlighted in Fig. 2A-F. These environments have characteristics that the policy does not experience during training. The terrains can deform and crumble, with significant variation of material properties over the surface. The robot’s legs are subjected to frequent disturbances due to vegetation, rubble, and sticky mud. Existing terrain estimation pipelines that use cameras or LiDAR [27] fail in environments with snow (Fig. 2A), water (Fig. 2C), or dense vegetation (Fig. 2F). Our controller does not rely on exteroception and is immune to such failure. The controller learns omnidirectional locomotion based on a history of proprioceptive observations and is robust in zero-shot deployment on terrains with characteristics that were never experienced during training.

We have compared the presented controller to a state-of-the-art baseline [1, 26] in the forest environment. The baseline could traverse flat and unobstructed patches, but failed frequently upon encountering loose branches, thick vegetation, and mud, as shown in Movie S1. Our controller never failed in these experiments.

We have quantitatively evaluated the presented controller and the baseline in three conditions: moss, mud, and vegetation (Fig. 2D-F). We have measured locomotion speed and energy efficiency. The results are reported in Table 1. The presented controller achieves higher locomotion speed in all conditions. We computed the dimensionless cost of transport (COT) to compare the efficiency of the controllers at different speed ranges. We define mechanical COT as $\sum_{12 \text{ actuators}} [\tau \dot{\theta}]^+ / (mgv)$. τ denotes joint torque, $\dot{\theta}$ is joint speed, mg is the total weight, and v is the locomotion speed. This quantity represents positive mechanical power exerted by the actuator per unit weight and unit locomotion speed [28]. As shown in Table 1, the presented controller is more energy-efficient, with a lower COT than the baseline.

The quantitative evaluation reported in Table 1 understates the difference between the two controllers because it only measures speed and energetic efficiency of the baseline when it successfully locomotes. The baseline’s catastrophic failures are not factored into these measurements: when the baseline fails, it is reset by a human operator in a more stable configuration. Catastrophic failures of the baseline controller due to thick vegetation and other factors are shown in Movie S1. Our controller exhibited no such failures.

DARPA Subterranean Challenge

Our controller was used by the CERBERUS team for the DARPA Subterranean Challenge Urban Circuit (Fig. 2G). It replaced a model-based controller that had been employed by the team in the past [1, 26]. The objective of the competition is to develop robotic systems that rapidly map, navigate, and search complex underground environments, including tunnels, urban underground, and cave networks. The human

operators are not allowed to assist the robots during the competition physically; only teleoperation is allowed. Accordingly, the locomotion controller needs to perform without failure over extended mission durations. To our knowledge, this is the first use of a legged locomotion controller trained via model-free RL in such competitive field deployment.

The presented controller drove two ANYmal-B robots in four missions of 60 minutes. The controller exhibited a zero failure rate throughout the competition. A steep staircase that was traversed by one of the robots during the competition is shown in Fig. 2G.

Indoor experiments

We further evaluated the robustness of the presented controller in an indoor environment populated by loose debris, as shown in Fig. 3A. Support surfaces are unstable and the robot’s feet frequently slip. Such conditions can be found at disaster sites and construction zones, where legged robots are expected to operate in the future.

Results are shown in Fig. 3A and Movie S2. The robot moves omnidirectionally over the area. The presented controller can stably locomote over shifting support surfaces. This level of robustness is beyond the reach of prior controllers for ANYmal robots [1, 26] and is comparable to the state of the art [2, 29].

The learned controller manifests a foot-trapping reflex, as shown in Fig. 3B and Movie S3. The policy identifies the trapping of the foot purely from proprioceptive observations and lifts the foot over the obstacle. Such reflexes were not specified in any way during training: they developed adaptively. This distinguishes the presented approach from conventional controller design methods, which explicitly build in such reflexes and orchestrate their execution by a higher-level state machine [1, 3]. The step shown in Fig. 3B is 16.8 cm high, which is higher than the foot clearance of the legs during normal walking on flat terrain. The maximum foot clearance on flat terrain is 12.9 cm and 13.6 cm for the LF and RF legs, respectively¹, and increases up to 22.5 cm and 18.5 cm in the case of foot-trapping. Our controller also learns to adapt the hind leg trajectories when stepping up. The maximum foot clearance on flat terrains is 13.5 cm and 9.06 cm for the LH and RH legs, and increases up to 16.6 cm and 15.9 cm when the front legs are above the step. Further analysis is provided in the Materials and Methods section. Note also that the reflexes learned by our controller are more general and are not tied to particular contact events. Fig. 3C shows the controller responding to a mid-shin collision during the swing phase. Here, the trapping event was not signalled by foot contact, and scripted controllers that use foot contact events as triggers would not appropriately handle this situation. Our controller, on the other hand, analyzes the proprioceptive stream as a whole and is trained without making assumptions about possible contact locations. Hence, it can learn to react to any obstructions and disturbances that impact the robot’s bodily configuration.

We now focus on comparing the presented approach with the baseline [1, 26] in controlled settings. We first compare the robustness of the controllers in the diagnostic setting of a single step, as shown in Fig. 3D. In each trial, the robot is driven straight to a step for 10 s. A trial is a success if the robot traverses the step with both front and hind legs. We conducted 10 trials for each step height and computed the success rate. Since the baseline controller takes a desired linear velocity of the base as input, we commanded a forward velocity of 0.2 m/s and 0.6 m/s. 0.6 m/s is the maximum speed of the baseline. The success rates are given in Fig. 3E. The presented controller outperforms the baseline in both stepping up and down. The baseline showed high sensitivity to foot-trapping, which often led to a fall, as shown in Movie S3.

¹We denote left, right, fore, and hind as L, R, F, H, respectively, to compactly refer to a leg. For example, ‘LF leg’ refers to the left fore leg.

数量	控制器	地形		
		Moss	Mud	植被
平均速度 (米/秒)	Ours	0.452	0.338	0.248
	基线	0.199	0.197	–
平均机械 COT	Ours	0.423	0.692	1.23
	基线	0.625	0.931	–

表1. 自然环境中运动性能的比较。 机械运输成本使用执行器输出的正机械功率计算。

山路、流动溪流、泥地、茂密植被、松散碎石、雪覆盖的山丘以及潮湿森林。图2A-F进一步突出了若干具体场景。这些环境具有策略在训练中未曾经历的特征。地形可能变形或碎裂，表面材料属性变化显著。由于植被、碎石和粘性泥地，机器人的腿部频繁受到扰动。现有的使用相机或激光雷达的地形估计管道 [27] 在雪地（图2A）、流水（图2C）或茂密植被（图2F）环境中会失效。我们的控制器不依赖外部感知，因此不受此类故障影响。该控制器基于本体感觉观测的历史记录学习全向运动，并在具有训练中从未经历的特征的地形上实现鲁棒的零样本部署。

我们将所提出的控制器与最先进基线进行了对比 [1, 26]，在森林环境中进行了测试。基线控制器能够穿越平坦且无障碍的区域，但在遇到散落的树枝、茂密植被和泥地时频繁失败，如电影S1所示。我们的控制器在这些实验中从未失败。

我们在三种条件下对提出的控制器和基线进行了定量评估：苔藓、泥地和植被（图2D-F）。我们测量了运动速度和能效。结果报告在表格1中。所提出的控制器在所有条件下均实现了更高的运动速度。我们计算了无量纲的运输成本（COT），以比较不同速度范围内控制器的效率。我们将机械运输成本定义为 $\sum_{12 \text{ actuators}} [\tau \dot{\theta}]^+ / (mgv)$ 。 τ 表示关节扭矩， $\dot{\theta}$ 是关节速度， mg 是总重量， v 是运动速度。该数量表示执行器每单位重量和单位运动速度所施加的正机械功率 [28]。如表格1所示，所提出的控制器能效更高，其COT低于基线控制器。

表格1中报告的定量评估低估了两种控制器之间的差异，因为它仅测量了基线控制器成功运动时的速度和能效。基线的灾难性故障并未计入这些测量中：当基线失败时，人类操作员会将其重置为更稳定的配置。基线控制器因茂密植被和其他因素导致的灾难性故障如电影S1所示。我们的控制器未出现此类故障。

DARPA地下挑战赛

我们的控制器被CERBERUS团队用于DARPA地下挑战赛城市赛段（图2G）。它取代了该团队过去使用的基于模型的控制 [1, 26]。比赛的目标是开发能够快速测绘、导航和搜索复杂地下环境（包括隧道、城市地下和洞穴网络）的机器人系统。人类

操作员在比赛期间不得物理协助机器人；仅允许远程操作。因此，运动控制器需要在长时间任务中无故障运行。据我们所知，这是首次在如此竞争性的现场部署中使用通过无模型强化学习训练的四足运动控制器。

所提出的控制器在四次60分钟的任务中驱动了两台 ANYmal-B机器人。该控制器在整个比赛中表现出零故障率。比赛中一台机器人穿越的陡峭楼梯如图2G所示。

室内实验

我们进一步在布满松散碎片的室内环境中评估了所提出控制器的鲁棒性，如图3A所示。支撑面不稳定，机器人的足部频繁打滑。这种条件常见于灾难现场和建筑工地，未来腿式机器人预计将在这些环境中运行。

结果如图3A和视频S2所示。机器人在该区域全向移动。所提出的控制器能够在变化的支撑面上稳定运动。这种鲁棒性水平超越了先前针对ANYmal机器人的控制器 [1, 26]，并且与当前最先进的技术水平相当 [2, 29]。

学习到的控制器表现出一种足部卡住反射，如图3B和视频S3所示。该策略纯粹通过本体感觉观测识别出足部被卡住，并将足部抬升越过障碍物。这种反射在训练过程中并未以任何方式指定：它们是自适应发展出来的。这使所提出的方法区别于传统控制器设计方法，后者会明确构建此类反射，并通过更高级别的状态机来编排其执行 [1, 3]。图3B中所示的台阶高度为16.8厘米，高于在平坦地形上正常行走时腿部的足部离地高度。在平坦地形上，左前腿和右前腿的最大足部离地高度分别为12.9厘米和13.6厘米¹，而在足部卡住的情况下，该高度分别增加至22.5厘米和18.5厘米。我们的控制器还学会了在上台阶时调整后腿轨迹。在平坦地形上，左后腿和右后腿的最大足部离地高度分别为13.5厘米和9.06厘米，当前腿位于台阶上方时，该高度分别增加至16.6厘米和15.9厘米。进一步的分析见材料与方法部分。另请注意，我们的控制器学习到的反射更具通用性，并不局限于特定的接触事件。图3C展示了控制器在摆动相期间对小腿中部碰撞的响应。在此情况下，卡住事件并非由足部接触信号指示，而使用足部接触事件作为触发器的脚本控制器将无法妥善处理这种情况。另一方面，我们的控制器将本体感觉流作为一个整体进行分析，并且训练过程中未对可能的接触位置做出假设。因此，它能够学会对任何影响机器人身体构型的障碍物和扰动做出反应。

我们现在重点在受控设置中比较所提出的方法与基线 [1, 26]。我们首先在单步的诊断设置中比较控制器的鲁棒性，如图3D所示。每次试验中，机器人被直线驱动至一个台阶，持续10秒。如果机器人用前腿和后腿都成功越过台阶，则该试验视为成功。我们对每个台阶高度进行了10次试验，并计算了成功率。由于基线控制器以期望的基座线速度作为输入，我们指令了0.2米/秒和0.6米/秒的前进速度。0.6米/秒是基线的最大速度。成功率如图3E所示。所提出的控制器在上台阶和下台阶方面均优于基线。基线对足部卡陷表现出高敏感性，这常常导致摔倒，如视频S3所示。

¹我们分别用L、R、F、H表示左、右、前、后，以简洁地指代某条腿。例如，“左前腿”即指左前腿。

We also tested the controllers in the presence of substantial model mismatch. We attached a 10 kg payload, as shown in Fig. 3D and Movie S4. This payload is 22.7 % of the total weight of the robot, and was never simulated during training. As shown in Fig. 3E, the presented controller can still traverse steps up to 13.4 cm despite the model mismatch. The baseline is incapable of traversing any steps under any command speed with the payload.

We then evaluate the tracking performance of the controllers on flat ground with the payload. We commanded each controller in 8 directions and measured the locomotion speed and the tracking error. Target speed is fixed to 0.4 m/s for the baseline controller, which is similar to the operating speed of the presented controller. In Fig. 3F, we show the velocity profiles of the controllers. Our controller locomotes at around 0.4 m/s in all directions and performs similarly with the payload. On the other hand, the locomotion speed of the baseline varies with direction, which can be seen by the anisotropic velocity profile, and the velocity profile shifts significantly off center with the payload. Fig. 3G shows the heading error of the controllers in each commanded direction. The heading error is the angle between the command velocity and the base velocity of the robot. The heading error of the presented controller is consistently smaller than the baseline, both with and without the payload. The baseline’s error in the lateral direction reaches $\sim 30^\circ$ and the baseline fails when a speed of (0.6 m/s) is commanded, as shown in Movie S4. In contrast, the average heading error of the presented controller stays within 10° with or without the payload. We conclude that the presented controller is much more robust to model mismatch.

Next we test robustness to foot slippage. To introduce slippage, we used a moistened whiteboard [1]. The results are shown in Movie S5. The baseline quickly loses balance, aggressively swings the legs, and falls. In contrast, the presented controller adapts to the slippery terrain and successfully locomotes in the commanded direction.

3. DISCUSSION

The presented results substantially advance the published state of the art in legged robotics. Beyond the results themselves, the methodology presented in this work can have broad applications. Prior to our work, a hypothesis could be held that training in simulation is fundamentally constrained by the limitations of simulation environments in representing the complexity of the physical world. Present-day technology is severely limited in its ability to simulate compliant contact, slippage, and deformable and crumbling terrain. As a result, phenomena such as mud, snow, thick vegetation, gushing water, and many others are beyond the capabilities of robotics simulation frameworks [30–32]. The sample complexity of model-free RL algorithms, which commonly require millions of time steps for training, further exacerbates the challenge by precluding reliance on frameworks that may require seconds of computation per time step.

Our work demonstrates that simulating the astonishing variety of the physical world may not be necessary. Our training environment features only rigid terrain, with no compliance or overground obstructions such as vegetation. Nevertheless, controllers trained in this environment successfully meet the diversity of field conditions encountered at deployment.

We see a number of limitations and opportunities for future work. First, the presented controller only exhibits the trot gait. This is narrower than the range of gate patterns discovered by quadrupeds in nature [33]. The gait pattern is constrained in part by the kinematics and dynamics of the robot, but the ANYmal machines are physically capable of multiple gates [26]. We hypothesize that training protocols and objectives that emphasize diversity can elicit these.

Second, the presented controller relies solely on proprioception.

This is a significant advantage in that the controller makes few assumptions on the sensor suite and is not susceptible to failure when exteroception breaks down. Indeed, existing work has argued that a blind (proprioceptive) controller should form the basis of a legged locomotion stack [3]. Nevertheless, blind locomotion is inherently limited. If the machine is commanded to walk off a cliff, it will. Even in less extreme conditions, the robot’s gait is fairly conservative since it must by necessity feel out the environment with its body as it locomotes. A major opportunity for future work is to use the presented methodology as a starting point in the development of a hybrid proprioceptive-exteroceptive controller that, like many animals, will be able to locomote even when vision and other external senses are disrupted, but will use exteroceptive data when it is provided. This will enable legged machines to autonomously traverse environments that may have fatal elements such as cliffs, and to raise speed and energetic efficiency in safer conditions.

More broadly, the presented results expedite the deployment of legged machines in environments that are beyond the reach of wheeled and tracked robots and are dangerous or inaccessible to humans, while the presented methodology opens new frontiers for training complex robotic systems in simulation and deploying them in the full richness and complexity of the physical world.

4. MATERIALS AND METHODS

A. Overview

The main objective of the presented controller is to locomote over rough terrain following a command. The command is given either by a human operator or by a higher-level navigation controller. In our formulation, unlike many existing works [12, 14, 16] that focus on tracking the target velocity of the base (${}^B_{IB}\hat{v}_T$), only the direction (${}^B_{IB}\hat{\phi}_T$) is given to the controller. The reason is that the feasible range of target speeds is often unclear on challenging terrain. For example, the robot can walk faster downhill than uphill.

The command vector is defined as $\langle ({}^B_{IB}\hat{\phi}_T)_{xy}, (\hat{\omega}_T)_z \rangle$. The first part is the target horizontal direction in base frame $({}^B_{IB}\hat{\phi}_T)_{xy} := \langle \cos(\psi_T), \sin(\psi_T) \rangle$, where ψ_T is the yaw angle to command direction in the base frame. The stop command is defined as (0.0, 0.0). The second part is the turning direction $(\hat{\omega}_T)_z \in \{-1, 0, 1\}$. 1 refers to counter-clockwise rotation along the base z-axis.

An overview of our method is given in Fig. 4. We use a privileged learning strategy inspired by “learning by cheating” [23] (Fig. 4A). We first train a teacher policy that has access to privileged information concerning the terrain. This teacher policy is then distilled into a proprioceptive student policy that does not rely on privileged information. The privileged teacher policy is confined to simulation, but the student policy is deployed on physical machines. One difference of our methodology from that of Chen et al. [23] is that we do not rely on expert demonstrations to train the privileged policy; rather, the teacher policy is trained via reinforcement learning.

The privileged teacher model is based on multi-layer perceptrons (MLPs) that receive information on the current state of the robot, properties of the terrain, and the robot’s contact with the terrain. The model computes a latent embedding $\bar{l}_t$ that represents the current state, and an action $\bar{a}_t$. The training objective rewards locomotion in prescribed directions.

After the teacher policy is trained, it is used to supervise a proprioceptive student policy. The student model is a temporal convolutional network (TCN) [22] that receives a sequence of N proprioceptive observations as input. The student policy is trained by imitation. The vectors $\bar{l}_t$ and $\bar{a}_t$ computed by the teacher policy are used to supervise the student. This is illustrated in Fig. 4A.

我们还在存在显著模型失配的情况下测试了控制器。我们附加了一个10公斤的负载，如图3D和视频S4所示。该负载占机器人总重量的22.7%，并且在训练过程中从未模拟过。如图3E所示，尽管存在模型失配，所提出的控制器仍然能够越过高达13.4厘米的台阶。而基线在带有负载的情况下，无论以何种指令速度都无法越过任何台阶。

随后，我们在平坦地面上评估了带负载情况下各控制器的跟踪性能。我们向每个控制器下达了8个方向的指令，并测量了运动速度与跟踪误差。基线控制器的目标速度固定为0.4米/秒，这与所提出控制器的工作速度相近。在图3F中，我们展示了各控制器的速度曲线。我们的控制器在所有方向上均能以约0.4米/秒的速度运动，且在带负载时表现相似。相比之下，基线控制器的运动速度随方向变化，这可以从各向异性速度曲线中看出，并且带负载时速度曲线明显偏离中心。图3G展示了每个指令方向上控制器的航向误差。航向误差是指令速度与机器人基座速度之间的夹角。所提出控制器的航向误差始终小于基线控制器，无论是否带负载。基线控制器在横向上的误差达到 $\sim 30^\circ$ ，并且当指令速度为（0.6米/秒）时，基线控制器失效，如电影S4所示。相比之下，所提出控制器的平均航向误差无论是否带负载均保持在 10° 以内。我们得出结论：所提出的控制器对模型失配具有更强的鲁棒性。

接下来，我们测试对足部打滑的鲁棒性。为了引入打滑，我们使用了一块湿润白板 [1]。结果如电影S5所示。基线方法很快失去平衡，剧烈摆动腿部并摔倒。相比之下，所提出的控制器能够适应湿滑地形，并成功沿指令方向移动。

3. 讨论

所呈现的结果显著推进了腿式机器人学领域的已发表技术现状。除了结果本身，本工作中提出的方法还具有广泛的应用前景。在我们工作之前，可能存在一种假设，即仿真训练从根本上受到仿真环境在表示物理世界复杂性方面的限制。当今技术在模拟柔顺接触、打滑以及可变形和易碎地形方面的能力严重受限。因此，诸如泥地、雪地、茂密植被、汹涌水流等现象超出了机器人仿真框架 [30–32] 的能力范围。无模型强化学习算法的样本复杂度（通常需要数百万个时间步进行训练）进一步加剧了这一挑战，因为它排除了对每时间步可能需要数秒计算量的框架的依赖。

我们的工作表明，模拟物理世界惊人的多样性可能并非必要。我们的训练环境仅包含刚性地形，没有柔顺性，也没有植被等地面障碍物。然而，在此环境中训练的控制器成功应对了部署时遇到的各种现场条件。

我们看到了若干局限性以及未来工作的机遇。首先，所提出的控制器仅表现出小跑步态。这比自然界中四足动物发现的步态模式范围更窄 [33]。步态模式部分受限于机器人的运动学和动力学，但ANYmal机器人在物理上能够实现多种步态 [26]。我们推测，强调多样性的训练协议和目标可以激发这些步态。

其次，所提出的控制器仅依赖本体感觉。

这是一个显著的优势，因为该控制器对传感器套件的假设很少，并且当外部感知失效时不易发生故障。事实上，已有研究认为，盲（本体感觉）控制器应构成腿式运动堆栈的基础 [3]。然而，盲运动本身存在局限性。如果机器人被指令走下悬崖，它就会照做。即使在不太极端的条件下，机器人的步态也相当保守，因为它必须通过身体与环境接触来感知环境。未来工作的一个重要机遇是，将所提出的方法作为开发混合本体-外部感觉控制器的起点。这种控制器将像许多动物一样，即使在视觉和其他外部感官受到干扰时也能运动，但在提供外部感觉数据时会加以利用。这将使腿式机器能够自主穿越可能存在悬崖等致命元素的环境，并在更安全的条件下提高速度和能量效率。

更广泛地说，所呈现的结果加速了腿式机器在轮式和履带式机器人无法触及、对人类危险或难以进入的环境中的部署，同时所提出的方法论为在仿真中训练复杂机器人系统并将其部署到丰富而复杂的物理世界中开辟了新的前沿。

4. 材料与方法

A. 概述

所提出的控制器的主要目标是让机器人按照指令在崎岖地形上移动。指令由人类操作员或高层导航控制器给出。在我们的公式中，与许多现有工作 [12, 14, 16] 侧重于跟踪基坐标系的目标速度 (${}^B_{IB}\hat{v}_T$) 不同，只有方向 (${}^B_{IB}\hat{\phi}_T$) 被提供给控制器。原因在于，在挑战性地形上，目标速度的可行范围往往不明确。例如，机器人下坡时可以比上坡时走得更快。

指令向量定义为 $\langle ({}^B_{IB}\hat{v}_T)_{xy}, (\hat{\omega}_T)_z \rangle$ 。第一部分是基坐标系中的目标水平方向 $({}^B_{IB}\hat{v}_T)_{xy} := \langle \cos(\psi_T), \sin(\psi_T) \rangle$ ，其中 ψ_T 是基坐标系中指令方向的偏航角。停止指令定义为 (0.0, 0.0)。第二部分是转向方向 $(\hat{\omega}_T)_z \in \{-1, 0, 1\}$ 。1 表示沿基坐标系 z 轴逆时针旋转。

图4概述了我们的方法。[23]（图4A）。我们采用了一种受“学习作弊”启发的特权学习策略。首先训练一个能够获取地形特权信息的教师策略。然后，该教师策略被蒸馏为一个不依赖特权信息的本体感受学生策略。特权教师策略仅限于仿真环境，而学生策略则部署在物理机器上。我们的方法与Chen等人的方法的一个区别在于，我们不依赖专家演示来训练特权策略；相反，教师策略是通过强化学习训练的。

特权教师模型基于多层感知机（MLPs），接收机器人当前状态、地形属性以及机器人与地形接触的信息。该模型计算一个表示当前状态的潜在嵌入 $\bar{l}_t$ ，以及一个动作 $\bar{a}_t$ 。训练目标奖励在指定方向上的运动。

在教师策略训练完成后，它被用于监督一个本体感受的学生策略。学生模型是一个时序卷积网络（TCN）[22]，该网络接收一个序列的 N 个本体感受观测作为输入。学生策略通过模仿进行训练。由向量 $\bar{l}_t$ 和 $\bar{a}_t$ （由教师策略计算得出）被用于监督学生。如图4A所示。

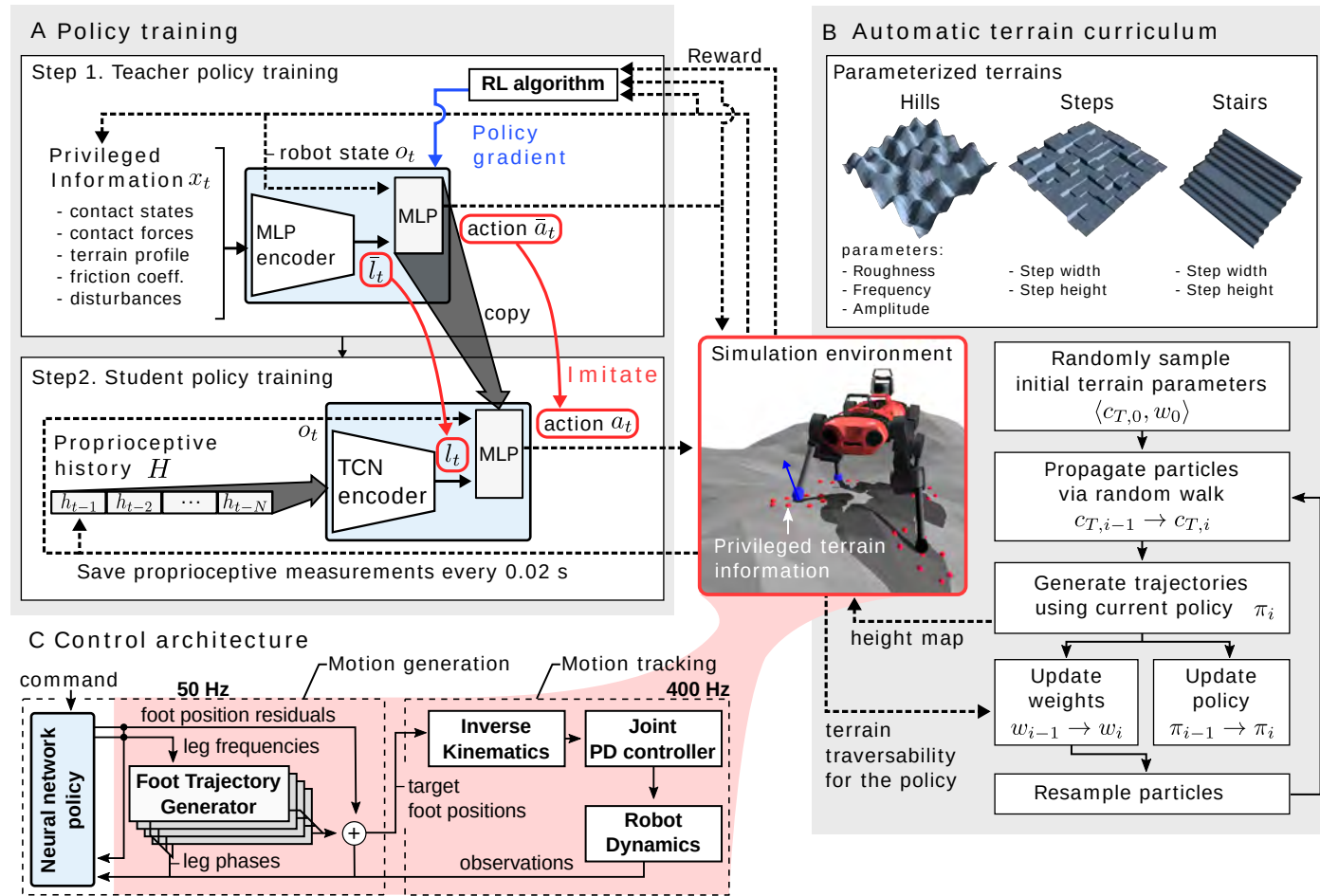


Fig. 4. Overview of the presented approach. (A) Two-stage training process. First, a teacher policy is trained using reinforcement learning in simulation. It has access to privileged information that is not available in the real world. Next, a proprioceptive student policy learns by imitating the teacher. The student policy acts on a stream of proprioceptive sensory input and does not use privileged information. (B) An adaptive terrain curriculum synthesizes terrains at an appropriate level of difficulty during the course of training. Particle filtering is used to maintain a distribution of terrain parameters that are challenging but traversable by the policy. (C) Architecture of the locomotion controller. The learned proprioceptive policy modulates motion primitives via kinematic residuals. An empirical model of the joint PD controller facilitates deployment on physical machines.

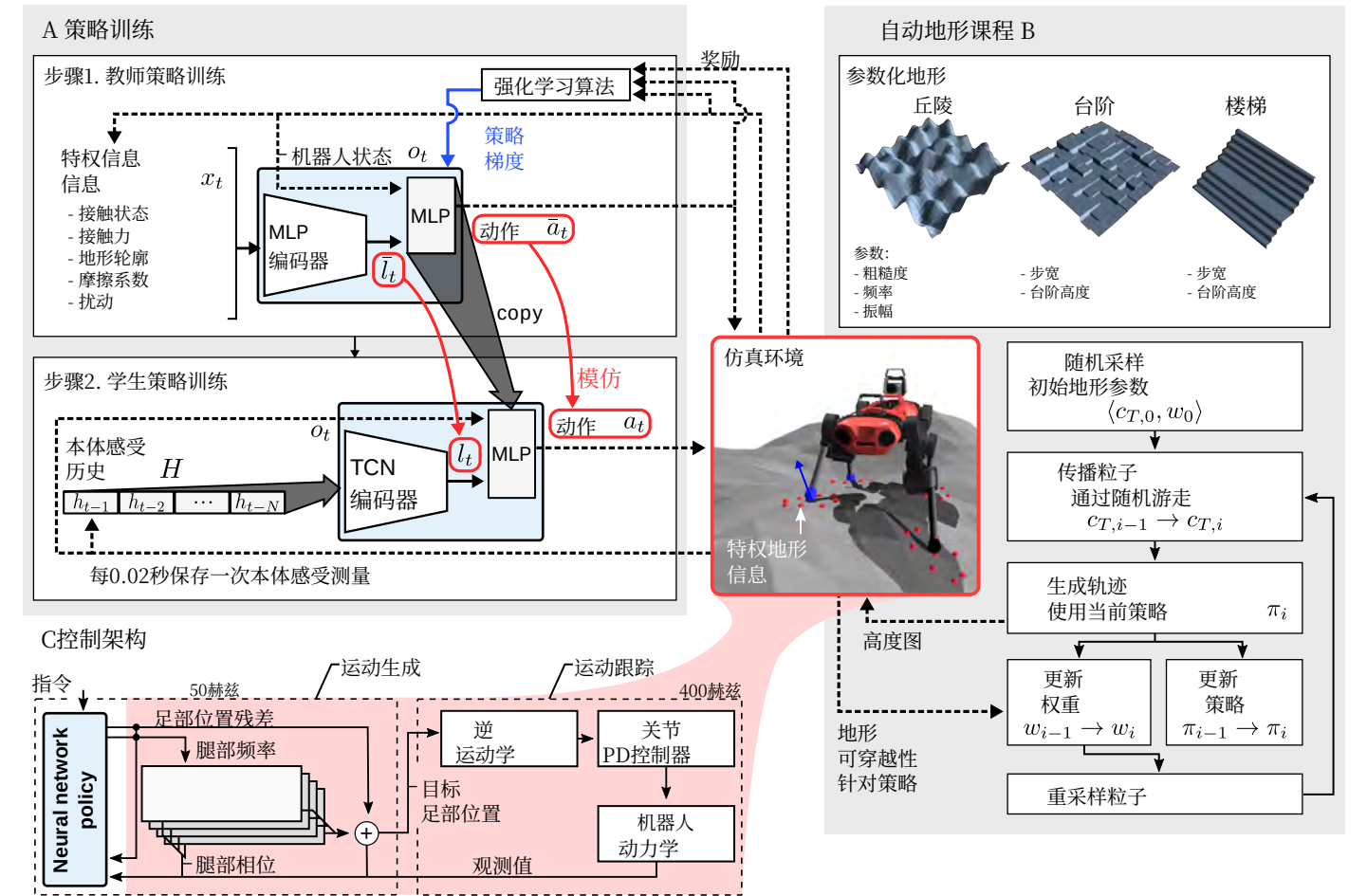


图 4. 所提出方法的概述。 (A) 两阶段训练过程。首先，在仿真中使用强化学习训练一个教师策略。该策略能够访问在现实世界中不可用的特权信息。接着，一个本体感受学生策略通过模仿教师进行学习。该学生策略基于本体感受感官输入流进行决策，不使用特权信息。 (B) 自适应地形课程在训练过程中合成难度适中的地形。粒子滤波用于维护一组具有挑战性但策略可通行的地形参数分布。 (C) 运动控制器的架构。学习得到的本体感受策略通过运动学残差来调制运动基元。关节PD控制器的经验模型有助于在物理机器上部署。

Training is conducted on procedurally generated terrains in simulation. The terrains are synthesized adaptively, to facilitate learning according to the skill level of the trained policies at any given time. We define a traversability measure for terrain and develop a sampling-based method to select terrains with the appropriate difficulty during the course of training. We use particle filtering to maintain an appropriate distribution of terrain parameters. This is illustrated in Fig. 4B. The terrain curriculum is applied during both teacher and student training.

Our control architecture is shown in Fig. 4C. We employ the Policies Modulating Trajectory Generators (PMTG) architecture [34] to provide priors on motion generation. The neural network policy modulates leg phases and motion primitives by synthesizing residual position commands.

The simulation uses a learned dynamics model of the robot’s joint PD controller [12]. This facilitates the transfer of policies from simulation to reality. After training in simulation, the proprioceptive controller is deployed directly on physical legged machines, with no fine-tuning.

Motion synthesis

We now elaborate on the control architecture that is illustrated in Fig. 4C. It is divided into motion generation and tracking. The input to our controller consists of the command vector and a sequence of proprioceptive measurements including base velocity, orientation and joint states. The controller does not use any exteroceptive input (e.g., no haptic sensors, cameras, or depth sensors). The input also does not contain any handcrafted features such as foot contact states or estimated terrain geometry. The controller outputs joint position targets.

Our motion generation strategy is based on the periodic leg phase. Previous works commonly leveraged predefined foot contact schedules [2, 26, 35]. We define a periodic phase variable $\phi_i \in [0.0, 2\pi)$ for each leg, which represents contact phase if $\phi \in [0.0, \pi)$ and swing phase if $\phi \in [\pi, 2\pi)$. At every time step t , $\phi_i = (\phi_{i,0} + (f_0 + f_i)t) \pmod{2\pi}$ where $\phi_{i,0}$ is the initial phase, f_0 is a common base frequency, and f_i is the frequency offset for the i -th leg. We want the legs to manifest periodic motions when $f_0 + f_i \neq 0$ and engage ground contact in contact phase. We set f_0 as 1.25 Hz, which is the value used by a previously developed conventional controller for a trot gait [26].

The target foot positions, which are the output of the motion generation block, are defined in the horizontal frames [35] of the feet ($H_i, i \in \{1, 2, 3, 4\}$). H_i is a reference frame that is attached below the hip joint of the i -th leg. The distance equals the nominal reach of the leg. The z-axis of the frame (${}^{H_i}z$) is parallel to e_g and ${}^{H_i}x$ is the projection of the base x-axis (Bx) onto the horizontal plane, i.e., the frame has the same yaw angle with the robot. The roll and pitch angles of H_i are decoupled from the base. This kinematic trick reduces the effect of base attitude on the foot motions [35] and consequently stabilizes training. Defining the output in H_i results in less premature termination at the beginning of the policy training, when the base motion is unstable due to random actions. Another benefit is that we can decompose the action distribution of the stochastic policy in the lateral and vertical directions during policy training. We applied larger noise in the lateral direction to promote exploration along the ground surface.

We use the PMTG architecture [34] to integrate a neural network to regulate the controller. Our implementation consists of four identical foot trajectory generators (FTGs) and a neural network policy. The FTG is a function $F(\phi) : [0.0, 2\pi) \rightarrow \mathbb{R}^3$ that outputs foot position targets for each leg. The FTG drives vertical stepping motion when f_i is non-zero. The definition of $F(\phi)$ is given in supplementary section S3.

The policy outputs f_i s and target foot position residuals ($\Delta r_{f_i,T}$), and the target foot position for the i -th foot is $r_{f_i,T} := F(\phi_i) + \Delta r_{f_i,T}$.

The tracking control is done using analytic inverse kinematics (IK) and joint position control. Each foot position target defined in H_i is

first expressed in the robot base frame, and the joint position targets are computed using analytic IK. The joint position targets are then tracked by joint position PD controllers. The main reason for using analytic IK is to maximize computational efficiency and to reuse existing position control actuator models [6, 15] for the sim-to-real transfer.

Teacher policy

We formulate the control problem as a Markov Decision Process (MDP). MDP is a mathematical framework for modeling discrete-time control processes in which the evolution of the state and the outcomes are partly stochastic. An MDP is defined by a state space $\mathcal{S}$, action space $\mathcal{A}$, a scalar reward function $\mathcal{R}(s_t, s_{t+1})$, and the transition probability $P(s_{t+1}|s_t, a_t)$. A learning agent selects an action a_t from its policy $\pi(a_t|s_t)$ and receives a reward r_t from the environment. The objective of the RL framework is to find an optimal policy π^* that maximizes the discounted sum of rewards over an infinite time horizon.

Assuming the environment is fully observable to the teacher, we formulate locomotion control as an MDP and use an off-the-shelf RL method [36] to solve it. In this section, we provide the MDP for teacher training, which is defined by a tuple of state space, action space, transition probability, and reward function.

The state is defined as $s_t := \langle o_t, x_t \rangle$, where o_t is the measurement vector obtainable from the robot and x_t is the privileged information that is usually not available in the real world. The detailed definitions are given in Table S4. o_t contains command, orientation, base twist, joint positions and velocities, ϕ_i s, f_i s, and previous foot position targets. Joint position errors and velocities measured at -0.01 s and -0.02 s are contained in o_t , which is the same as the input to the learned model of the joint-level PD controller. This information allows the policy to exploit the actuator dynamics [12]. To encode the leg phase, we use $\langle \cos(\phi), \sin(\phi) \rangle$ instead of ϕ , which is a smooth and unique representation for the angle. Previous foot position targets are also fed back to the policy and are used to compute the target smoothness reward that is explained in the following paragraph. When the student controller is deployed, the quantities in o_t are replaced with readings from the proprioceptive sensors and the base velocity and orientation are provided by a state estimator [37]. x_t contains noiseless information that we receive directly from a physics engine. x_t mainly consists of information related to foot-ground interactions such as terrain profile, foot contact states and forces, friction coefficients, and external disturbance forces applied during training. Specifically, we represent the terrain profile with the elevation of 9 scan points around each foot, which are symmetrically placed along a circle with a 10 cm radius (visualized in Fig. 4).

The action ($\bar{a}_t$) is a 16-dimensional vector consisting of leg frequencies and foot position residuals.

The reward function is defined such that an RL agent receives a higher reward if it advances faster towards the goal. The reward function is specified in detail in supplementary section S4.

The policy network is constructed by two MLP blocks as shown in Fig. 4A. The MLP encoder embeds x_t into a latent vector $\bar{l}_t$. The command and robot states are not included in x_t , so $\bar{l}_t$ contains only the terrain- and contact-related features. We hypothesize that $\bar{l}_t$ drives adaptive behaviors such as changing foot clearance depending on the terrain profile. Then $\bar{l}_t$ and o_t are provided to the subsequent MLP layers to compute action.

The Trust Region Policy Optimization (TRPO) [36] algorithm is used for training. The hyperparameters we used are given in Table S7.

Student policy

The proprioceptive student policy only has access to o_t . A key hypothesis here is that the latent features $\bar{l}_t$ can be (partially) recovered from

训练在仿真中的程序化生成地形上进行。地形通过自适应方式合成，以便根据训练策略在任何给定时间点的技能水平促进学习。我们定义了一种地形的可穿越性度量，并开发了一种基于采样的方法，在训练过程中选择适当难度的地形。我们使用粒子滤波来维持地形参数的适当分布。如图4B所示。地形课程在教师训练和学生训练期间均会应用。

我们的控制架构如图4C所示。我们采用策略调制轨迹生成器架构 [34] 来为运动生成提供先验知识。神经网络策略通过合成残差位置指令来调制腿部相位和运动基元。

仿真使用了机器人关节PD控制器的学习动力学模型 [12]。这有助于策略从仿真迁移到现实。在仿真中完成训练后，本体感受控制器可直接部署在物理腿式机器上，无需微调。

运动合成

我们现在详细阐述图4C所示的控制架构。该架构分为运动生成和跟踪两部分。我们控制器的输入包括指令向量以及一系列本体感受测量值，如基座速度、朝向和关节状态。控制器不使用任何外部感受输入（例如，没有触觉传感器、相机或深度传感器）。输入也不包含任何手工设计的特征，如足部接触状态或估计的地形几何。控制器的输出是关节位置目标。

我们的运动生成策略基于周期性腿相位。以往的工作通常利用预定义的足部接触时间表 [2, 26, 35]。我们为每条腿定义一个周期性相位变量 $\phi_i \in [0.0, 2\pi)$ ，当 $\phi \in [0.0, \pi)$ 时表示接触相位，当 $\phi \in [\pi, 2\pi)$ 时表示摆动相。在每个时间步 t ， $\phi_i = (\phi_{i,0} + (f_0 + f_i)t) \pmod{2\pi}$ ，其中 $\phi_{i,0}$ 是初始相位， f_0 是公共基频，而 f_i 是第 i 条腿的频率偏移。我们希望当 $f_0 + f_i \neq 0$ 时，腿部表现出周期性运动，并在接触相位中与地面接触。我们将 f_0 设置为 1.25 Hz，这是先前为小跑步态开发的传统控制器所使用的值 [26]。

目标足位置（即运动生成模块的输出）定义在足部的水平坐标系 [35] 中 ($H_i, i \in \{1, 2, 3, 4\}$)。 H_i 是一个参考坐标系，位于第 i 条腿的髋关节下方。该距离等于腿的标称可达范围。该坐标系的 z 轴 (${}^{H_i}z$) 平行于 e_g ，而 ${}^{H_i}x$ 是基座 x 轴 (Bx) 在水平面上的投影，即该坐标系与机器人具有相同的偏航角。 H_i 的横滚角和俯仰角与基座解耦。这一运动学技巧减少了基座姿态对足部运动 [35] 的影响，从而稳定了训练。在策略训练初期，当基座运动因随机动作而不稳定时，将输出定义在 H_i 中可减少过早终止的情况。另一个好处是，我们可以在策略训练期间将随机策略的动作分布分解为横向和垂直方向。我们在横向方向上施加了更大的噪声，以促进沿地面的探索。

我们采用 PMTG 架构 [34] 将神经网络集成到控制器中。我们的实现包含四个相同的足部轨迹生成器 (FTGs) 和一个神经网络策略。FTG 是一个函数 $F(\phi) : [0.0, 2\pi) \rightarrow \mathbb{R}^3$ ，为每条腿输出足部位置目标。当 f_i 非零时，FTG 驱动垂直步态运动。 $F(\phi)$ 的定义见补充章节 S3。

策略输出 f_i s 和目标足部位置残差 ($\Delta r_{f_i,T}$)，第 i 条腿的目标足部位置为 $r_{f_i,T} := F(\phi_i) + \Delta r_{f_i,T}$ 。跟踪控制通过解析逆运动学 (IK) 和关节位置控制实现。每个在 H_i 中定义的足位置目标

首先在机器人基坐标系中表示，然后通过解析逆运动学计算关节位置目标。关节位置目标随后由关节位置 PD 控制器进行跟踪。采用解析逆运动学的主要原因是最大化计算效率，并复用现有的位置控制执行器模型 [6, 15] 以支持仿真到现实迁移。

教师策略

我们将控制问题建模为马尔可夫决策过程 (MDP)。MDP 是一种用于建模离散时间控制过程的数学框架，其中状态演化和结果部分具有随机性。一个 MDP 由状态空间 $\mathcal{S}$ 、动作空间 $\mathcal{A}$ 、标量奖励函数 $\mathcal{R}(s_t, s_{t+1})$ 以及转移概率 $P(s_{t+1}|s_t, a_t)$ 定义。学习智能体从其策略 $\pi(a_t|s_t)$ 中选择动作 a_t ，并从环境中获得奖励 r_t 。强化学习框架的目标是找到最优策略 π^* ，以最大化无限时间范围内的折扣奖励总和。

假设环境对教师完全可观测，我们将运动控制建模为 MDP，并使用现成强化学习方法 [36] 求解。本节将给出教师训练的 MDP 定义，该定义由状态空间、动作空间、转移概率和奖励函数的元组构成。

状态定义为 $s_t := \langle o_t, x_t \rangle$ ，其中 o_t 是从机器人获取的测量向量， x_t 是现实世界中通常不可获取的特权信息。详细定义见表 S4。 o_t 包含指令、朝向、基座扭转、关节位置与速度、 ϕ_i s、 f_i s 以及先前足部位置目标。 o_t 中包含在 -0.01 秒和 -0.02 秒时测量的关节位置误差和速度，这与关节级 PD 控制器学习模型的输入相同。这些信息使策略能够利用执行器动力学 [12]。为编码腿部相位，我们使用 $\langle \cos(\phi), \sin(\phi) \rangle$ 替代 ϕ ，这是对角度的一种平滑且唯一的表示。先前足部位置目标也会反馈给策略，用于计算下文将解释的目标平滑度奖励。当学生控制器部署时， o_t 中的量被替换为本体感觉传感器的读数，而基座速度和朝向则由状态估计器 [37] 提供。 x_t 包含直接从物理引擎获取的无噪声信息。 x_t 主要包含与足地交互相关的信息，如地形轮廓、足部接触状态与力、摩擦系数以及训练过程中施加的外部干扰力。具体而言，我们通过每只足周围 9 个扫描点的高程来表示地形轮廓，这些扫描点沿半径为 10 厘米的圆对称分布（如图 4 所示）。

动作 ($\bar{a}_t$) 是一个 16 维向量，由腿部频率和足部位置残差组成。

奖励函数的定义是：强化学习智能体在更快地向目标前进时获得更高的奖励。奖励函数的详细说明见补充材料第 S4 节。

策略网络由两个多层感知机块构成，如图 4A 所示。多层感知机编码器将 x_t 嵌入到一个潜在向量 $\bar{l}_t$ 中。指令和机器人状态不包含在 x_t 中，因此 $\bar{l}_t$ 仅包含地形和接触相关的特征。我们假设 $\bar{l}_t$ 驱动自适应行为，例如根据地形的轮廓改变足部离地高度。然后 $\bar{l}_t$ 和 o_t 被提供给后续的多层感知机层以计算动作。

训练采用信任区域策略优化 (TRPO) [36] 算法。我们使用的超参数见表 S7。

学生策略

本体感受学生策略仅能访问 o_t 。这里的一个关键假设是，潜在特征 $\bar{l}_t$ 可以（部分地）从

a time series of proprioceptive observations, h_t , which is defined as $h_t := o_t \setminus \{f_o, \text{joint history, previous foot position targets}\}$.

The student policy uses a temporal convolutional network (TCN) [22] encoder. The input to the TCN encoder is $H = \{h_{t-1}, \dots, h_{t-N-1}\}$, where N is the history length. The encoder is fully convolutional and consists of three dilated causal convolutional layers, interleaved with strided convolutional layers that reduce dimensionality. The architecture is specified in Tables S5 and S6.

We use the TCN architecture because it affords transparent control over the input history length, can accommodate long histories, and is known to be robust to hyperparameter settings [22]. A comparison with a recurrent neural network architecture is provided in supplementary section S8.

The student policy is trained via supervised learning. The loss function is defined as

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t, H))^2 + (\bar{l}_t(o_t, x_t) - l_t(H))^2. \quad (1)$$

Quantities marked by a bar ($\bar{\cdot}$) denote target values generated by the teacher. We employ the dataset aggregation strategy (Dagger) [38]. Specifically, training data is generated by rolling out trajectories by the student policy. For each visited state, the teacher policy computes its embedding and action vectors ($\bar{\cdot}$). These outputs of the teacher policy are used as supervisory signals associated with the corresponding states. The hyperparameters we used are given in Table S8.

Adaptive terrain curriculum

Our method is inspired by automatic curriculum learning (ACL) for RL agents [25, 39]. The paired open-ended trailblazer (POET) approach [25] generates diverse parameterized terrains for a 2D bipedal agent. The method employs minimal criteria (MC) [24, 40] and aims to choose environmental parameters that are neither too challenging nor trivial for the agents: this is realized by selecting task parameters that yield mid-range rewards. Florensa et al. [39] similarly choose achievable yet difficult goals for RL agents.

Our method likewise realizes a training curriculum that gradually modifies a distribution over environmental parameters such that the policy can continuously improve locomotion skills and generalize to new environments. Our work differs from POET as POET aims for open-ended search in the space of possible problems and evolves a population of specialized agents while we seek to obtain a single generalist agent.

Fig. 4B shows the types of terrains used in our training environment. Each terrain is generated by a parameter vector $c_T \in \mathcal{C}$. The terrains are described in detail in supplementary section S5. Our ACL method approximates a distribution of desirable c_T s using a particle filter.

We first describe how a given c_T is evaluated in simulation. Instead of directly using the reward function to evaluate the learning progress [25, 41–43], we evaluate c_T s by the traversability of generated terrains, which is defined as the success rate of traversing a terrain. We found traversability to be more intuitive than the reward function, which consists of multiple objectives that are often unbounded. We first define a labeling function ν as

$$\nu(s_t, a_t, s_{t+1}) = \begin{cases} 1 & \text{if } v_{pr}(s_{t+1}) > 0.2 \\ 0 & \text{if } v_{pr}(s_{t+1}) < 0.2 \vee \text{termination} \end{cases} \quad (2)$$

for a state transition from s_t to s_{t+1} . $v_{pr}(s_{t+1})$ stands for the inner product of the base velocity and commanded direction at time step $t+1$. If π can locomote in the commanded direction faster than 0.2 m/s, we consider the terrain traversable in this direction. The threshold is a hyperparameter; 0.2 m/s is about one third of the maximum speed of our robot. Traversability is defined as

$$Tr(c_T, \pi) = \mathbb{E}_{\xi \sim \pi} \{\nu(s_t, a_t, s_{t+1} \mid c_T)\} \in [0.0, 1.0], \quad (3)$$

where ξ refers to trajectories generated by π . This follows a definition of empirical traversability in prior work [44].

The objective of our terrain generation method is to find c_T s with mid-range traversability ($Tr(c_T, \pi) \in [0.5, 0.9]$). The rationale is to synthesize terrains that are neither too easy nor too difficult. We define terrain desirability as follows:

$$Td(c_T, \pi) := \Pr(Tr(c_T, \pi) \in [0.5, 0.9]) \quad (4)$$

$$= \mathbb{E}_{\xi \sim \pi} \{Tr(c_T, \pi) \in [0.5, 0.9]\}, \quad (5)$$

where 0.5 and 0.9 are fixed thresholds for minimum/maximum traversability.

We use a particle filter to keep track of a distribution of high-desirability c_T s during training. We formulate a particle filtering problem where we approximate the distribution of terrain parameters that satisfies $Tr(c_T, \pi) \in [0.5, 0.9]$ with a finite set of sampling points ($c_T^k \in \mathcal{C}, k \in 1, \dots, N_{particle}$). Our algorithm is modeled on the Sequential Importance Resampling (SIR) particle filter. It is based on the following assumptions.

1. Terrain parameters with similar $Tr(\cdot, \pi)$ are close in Euclidean distance in parameter space.
2. A policy trained over the terrains generated by c_T s in some area of $\mathcal{C}$ will learn to interpolate to nearby terrain parameters.
3. $c_{T,0}, c_{T,1}, \dots$ forms a Markov process, where $c_{T,j} = \{c_{T,j}^1, c_{T,j}^2, \dots, c_{T,j}^{N_{particle}}\}$ at iteration j .

The first assumption comes from the insight that terrain parameters can be interpolated, e.g., the difficulty of a staircase increases as we increase the step height. The second assumption justifies the use of discrete samples from $\mathcal{C}$ to train a policy that generalizes over a certain region of $\mathcal{C}$. The last assumption is necessary for formulating a particle filter.

The importance weight w^k is defined for each c_T^k , and the set of tuples $\langle c_T^k, w^k \rangle$ approximates the target distribution (c_T s with $Tr(c_T, \pi) \in [0.5, 0.9]$). We define the measurement variable y_j^k such that $y_j^k = 1$ if $Tr(c_{T,j}^k, \pi) \in [0.5, 0.9]$. Then the terrain desirability defined above becomes the measurement probability

$$\Pr(y_j^k | c_{T,j}^k) = \Pr(Tr(c_{T,j}^k, \pi) \in [0.5, 0.9]) = Td(c_{T,j}^k, \pi). \quad (6)$$

For practical implementation, the measurement probability is computed by the empirical expectation from the samples collected during policy training:

$$\Pr(y_j^k | c_{T,j}^k) \approx \sum \frac{\mathbb{1}(Tr(c_{T,j}^k, \pi) \in [0.5, 0.9])}{N_{traj}}, \quad (7)$$

where N_{traj} denotes the number of trajectories generated using $c_{T,j}^k$. The trajectories are also used for policy training. Our method therefore does not require additional evaluation steps to advance the curriculum of the terrain parameters. Resampling is done such that the probability of choosing the k th sample equals the normalized importance weight $w^k / \sum_i^{N_{particle}} w^i \in [0, 1]$.

The transition model is a random walk in $\mathcal{C}$. Each parameter of a sampling point is shifted to its adjacent value by a fixed probability $p_{transition}$. It satisfies the third assumption (Markov process) because the evolution of each parameter only relies on the current value and randomly sampled noise. To improve exploration, we bounded and discretized $\mathcal{C}$ to reduce the search space. The initial samples ($c_{T,0}^k$) are either drawn uniformly from $\mathcal{C}$ or concentrated on almost flat terrains.

Implementation details and an overview of the training process are provided in supplementary section S2 and Algorithm S1 in the supplement.

本体感觉观测值的时间序列 h_t 中恢复，该序列定义为 $h_t := o_t \setminus \{f_o, \text{关节历史、先前足部位置目标}\}$ 。

学生策略使用时序卷积网络（TCN）[22]编码器。TCN编码器的输入是 $H = \{h_{t-1}, \dots, h_{t-N-1}\}$

其中 N 是历史长度。该编码器为全卷积结构，由三个扩张因果卷积层与步进卷积层交错组成，后者用于降低维度。该架构详见表S5和表S6。

我们采用TCN架构，因为它能够透明地控制输入历史长度，可容纳长历史序列，并且已知对超参数设置具有鲁棒性[22]。与循环神经网络架构的比较见补充章节S8。

学生策略通过监督学习进行训练。损失函数定义为

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t, H))^2 + (\bar{l}_t(o_t, x_t) - l_t(H))^2. \quad (1)$$

带有横线标记的量 ($\bar{\cdot}$) 表示由教师生成的目标值。我们采用数据集聚合策略（Dagger）[38]。具体而言，训练数据通过学生策略展开轨迹生成。对于每个访问到的状态，教师策略计算其嵌入和动作向量 ($\bar{\cdot}$)。教师策略的这些输出被用作对应状态的监督信号。我们使用的超参数见表S8。

自适应地形课程

我们的方法受到针对强化学习智能体的自动课程学习（ACL）的启发[25, 39]。配对开放式开拓者（POET）方法[25]为二维双足智能体生成了多样化的参数化地形。该方法采用最小准则（MC）[24, 40]，旨在选择既不过于困难也不过于简单的环境参数：这是通过选择能产生中等范围奖励的任务参数来实现的。Florensa等人[39]同样为强化学习智能体选择了可实现但具有挑战性的目标。

我们的方法同样实现了一个训练课程，该课程逐步修改环境参数的分布，从而使策略能够持续提升运动技能并泛化到新环境。我们的工作与POET不同，因为POET旨在可能的问题空间中进行开放式搜索，并演化出一个专门化智能体的种群，而我们则寻求获得一个单一的通用智能体。

图4B展示了训练环境中使用的地形类型。每个地形由参数向量 $c_T \in \mathcal{C}$ 生成。这些地形在补充材料第S5节中有详细描述。我们的ACL方法使用粒子滤波器来近似理想 c_T 的分布。

我们首先描述如何在仿真中评估给定的 c_T 。我们不直接使用奖励函数来评估学习进度[25, 41–43]，而是通过生成地形的可穿越性来评估 c_T ，该可穿越性定义为穿越地形的成功率。我们发现可穿越性比奖励函数更直观，因为奖励函数包含多个通常无界的优化目标。我们首先定义一个标注函数 ν 如下：

$$\nu(s_t, a_t, s_{t+1}) = \begin{cases} 1 & \text{if } v_{pr}(s_{t+1}) > 0.2 \\ 0 & \text{if } v_{pr}(s_{t+1}) < 0.2 \vee \text{termination} \end{cases} \quad (2)$$

对于从 s_t 到 s_{t+1} 的状态转移， $v_{pr}(s_{t+1})$ 表示在时间步 $t+1$ 时基座速度与指令方向的内积。如果 π 能够以超过0.2米/秒的速度沿指令方向移动，则认为该方向上的地形可通行。该阈值是一个超参数；0.2米/秒约为我们机器人最大速度的三分之一。可穿越性定义为

$$Tr(c_T, \pi) = \mathbb{E}_{\xi \sim \pi} \{\nu(s_t, a_t, s_{t+1} \mid c_T)\} \in [0.0, 1.0], \quad (3)$$

其中 ξ 指由 π 生成的轨迹。这遵循了先前工作[44]中关于经验可通行性的定义。

我们的地形生成方法的目标是找到具有中等范围可通行性（ $Tr(c_T, \pi) \in [0.5, 0.9]$ ）的 c_T 。其原理是合成既不太简单也不太困难的地形。我们将地形期望度定义如下：

$$Td(c_T, \pi) := \Pr(Tr(c_T, \pi) \in [0.5, 0.9]) \quad (4)$$

$$= \mathbb{E}_{\xi \sim \pi} \{Tr(c_T, \pi) \in [0.5, 0.9]\}, \quad (5)$$

其中0.5和0.9是用于最小/最大可穿越性的固定阈值。

我们使用粒子滤波器在训练期间跟踪高期望度 c_T 的分布。我们构建了一个粒子滤波问题，其中我们通过一组有限的采样点（ $c_T^k \in \mathcal{C}, k \in 1, \dots, N_{particle}$ ）来近似满足 $Tr(c_T, \pi) \in [0.5, 0.9]$ 的地形参数分布。我们的算法基于序列重要性重采样（SIR）粒子滤波器建模，并基于以下假设。

1. 地形参数中， $Tr(\cdot, \pi)$ 相似的地形在参数空间中的欧几里得距离相近。
2. 在 $\mathcal{C}$ 的某个区域内，由 c_T s生成的地形上训练的策略，将学会插值到附近的地形参数。
3. $c_{T,0}, c_{T,1}, \dots$ 构成一个马尔可夫过程，其中 $c_{T,j} = \{c_{T,j}^1, c_{T,j}^2, \dots, c_{T,j}^{N_{particle}}\}$ 在第 j 次迭代。

第一个假设源于地形参数可以插值的洞察，例如，随着台阶高度的增加，楼梯的难度也随之增加。第二个假设证明了使用来自 $\mathcal{C}$ 的离散样本来训练一个在 $\mathcal{C}$ 的某个区域上具有泛化能力的策略是合理的。最后一个假设对于构建粒子滤波器是必要的。

重要性权重 w^k 为每个 c_T^k 定义，元组集合 $\langle c_T^k, w^k \rangle$ 近似于目标分布（ c_T s满足 $Tr(c_T, \pi) \in [0.5, 0.9]$ ）。我们定义测量变量 y_j^k ，使得 $y_j^k = 1$ 如果 $Tr(c_{T,j}^k, \pi) \in [0.5, 0.9]$ 。那么上述定义的地形期望度就变成了测量概率。

$$\Pr(y_j^k | c_{T,j}^k) = \Pr(Tr(c_{T,j}^k, \pi) \in [0.5, 0.9]) = Td(c_{T,j}^k, \pi). \quad (6)$$

在实际实现中，测量概率通过策略训练期间收集的样本的经验期望来计算：

$$\Pr(y_j^k | c_{T,j}^k) \approx \sum \frac{\mathbb{1}(Tr(c_{T,j}^k, \pi) \in [0.5, 0.9])}{N_{traj}}, \quad (7)$$

其中 N_{traj} 表示使用 $c_{T,j}^k$ 生成的轨迹数量。这些轨迹也用于策略训练。因此，我们的方法不需要额外的评估步骤来推进地形参数的课程。重采样的方式是，选择第 k 个样本的概率等于归一化重要性权重 $w^k / \sum_i^{N_{particle}} w^i \in [0, 1]$ 。

转移模型是 $\mathcal{C}$ 中的随机游走。采样点的每个参数以固定概率 $p_{transition}$ 偏移到其相邻值。这满足第三个假设（马尔可夫过程），因为每个参数的演化仅依赖于当前值和随机采样的噪声。为了改善探索，我们对 $\mathcal{C}$ 进行了边界限定和离散化，以缩小搜索空间。初始样本（ $c_{T,0}^k$ ）要么从 $\mathcal{C}$ 中均匀采样，要么集中在几乎平坦的地形上。

实现细节和训练过程的概述见补充材料第S2节和补充材料中的算法S1。

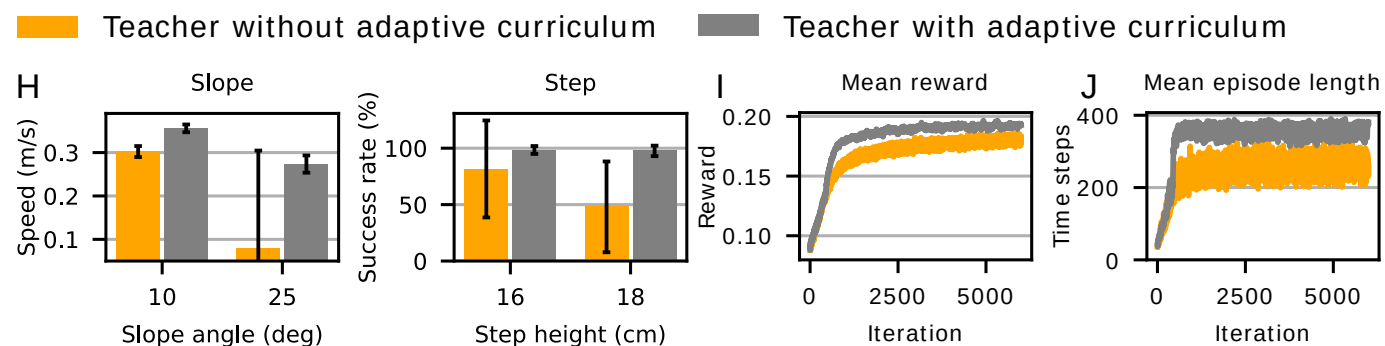
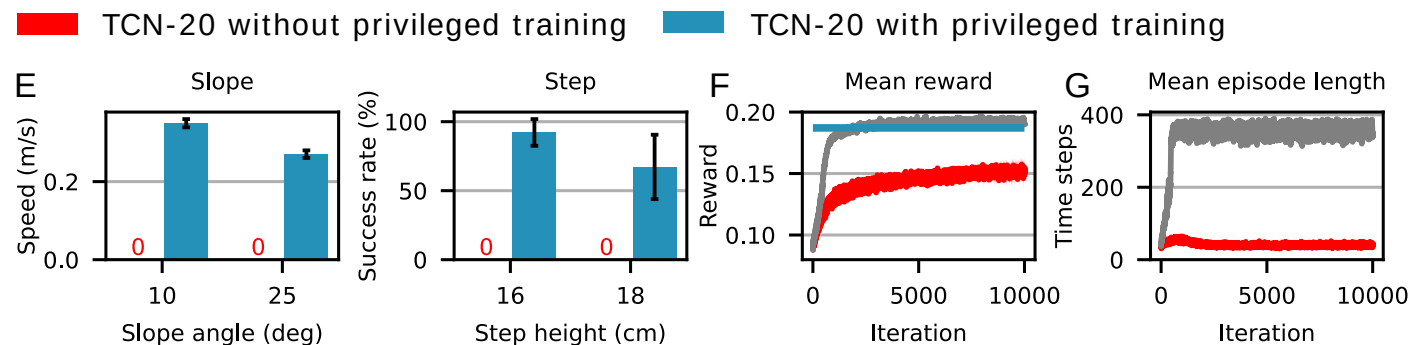
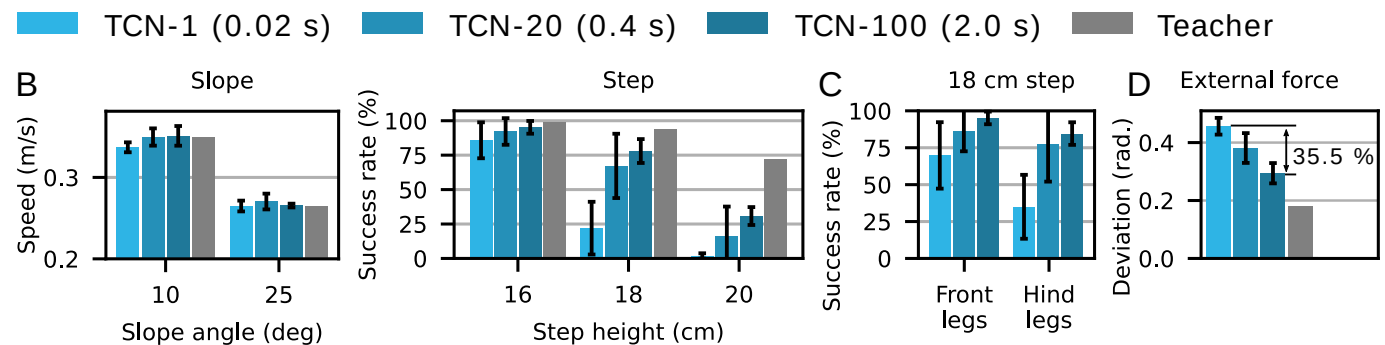
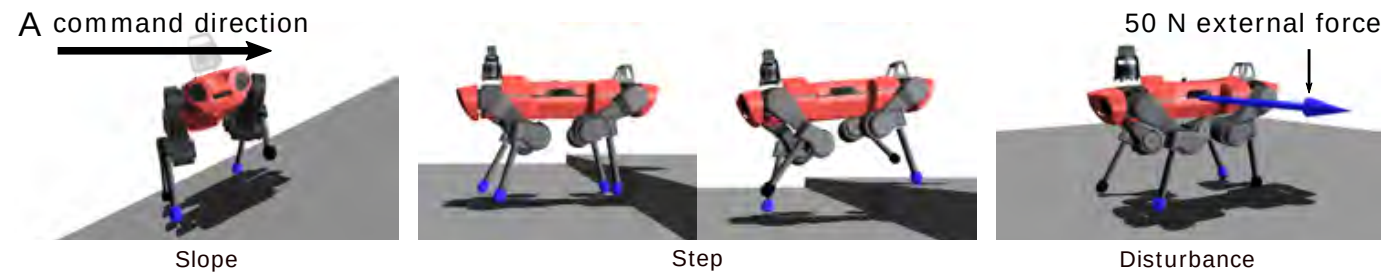


Fig. 5. Ablation studies. We trained each model 5 times using different random seeds. Error bars denote 95 % confidence intervals. (A) Test setups. The robot is commanded to advance for 10 s in the specified direction (black arrow). We conducted 100 trials for each test. On the step test, a trial is considered successful if the robot traverses the step with both front and hind legs. Robots are initialized with random joint configurations. Initial yaw angle is sampled from $U(-\pi, \pi)$ for the slope test and from $U(-\pi/6, \pi/6)$ for the other tests. The friction coefficients between the feet and the ground are sampled from $U(0.4, 1.0)$. The external force is applied for 5 s in the lateral direction. (B-D) Importance of memory length N in the TCN- N encoder. (E-G) Importance of privileged training. (F) Learning curves for the teacher (grey) and a TCN-20 student trained directly, without privileged training (red). For comparison, the blue line indicates the mean reward of a TCN-20 student trained with privileged training. The reward is computed by running each policy on uniformly sampled terrains. (H-J) Importance of the adaptive curriculum.

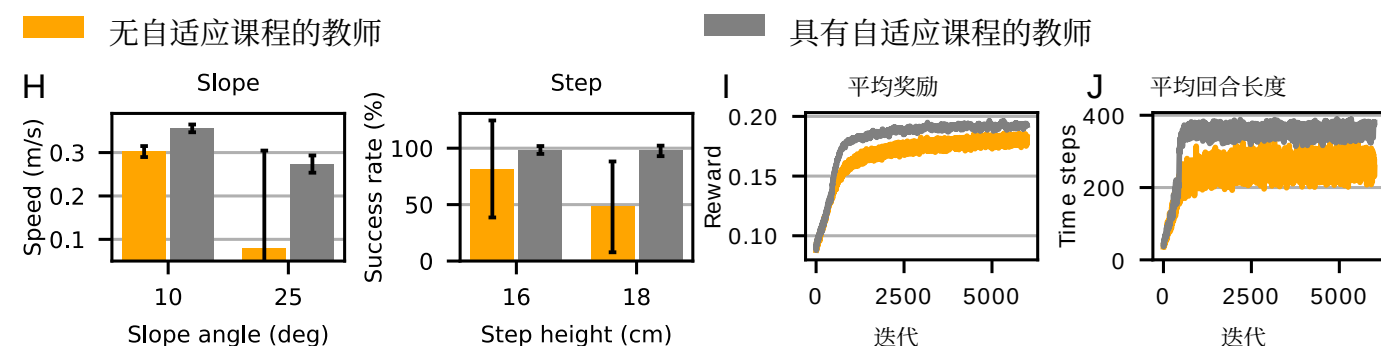
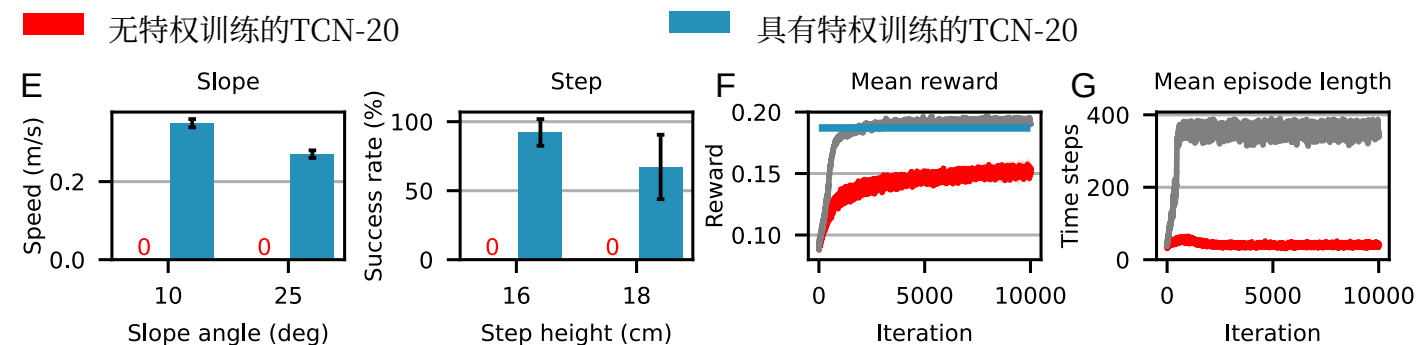
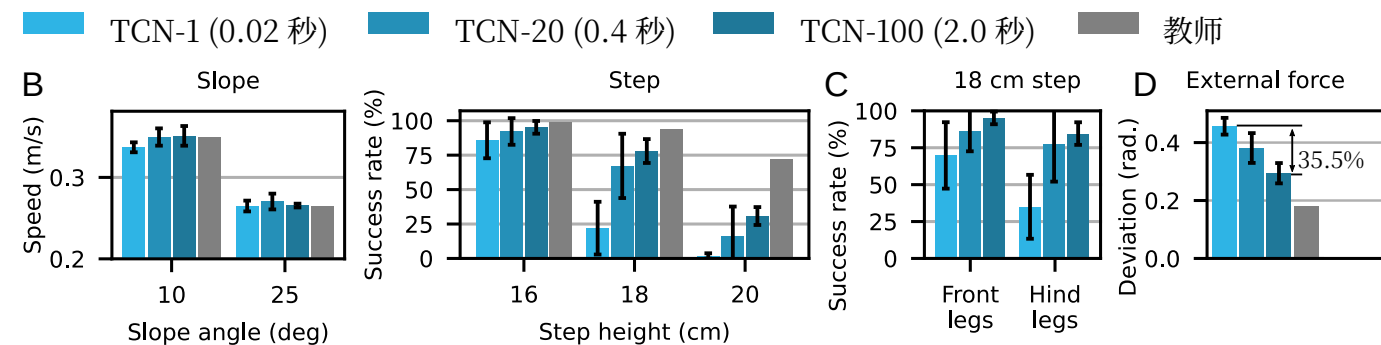
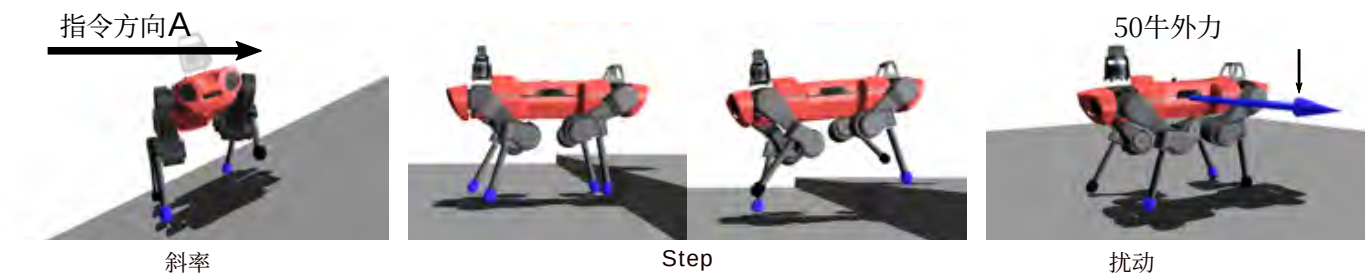


图5. 消融研究. 我们使用不同的随机种子对每个模型进行了5次训练。误差条表示95%置信区间。(A) 测试设置。机器人被指令沿指定方向（黑色箭头）前进10秒。我们对每个测试进行了100次试验。在台阶测试中，如果机器人用前腿和后腿都越过台阶，则该试验被视为成功。机器人以随机关节配置初始化。斜坡测试的初始偏航角从 $U(-\pi, \pi)$ 中采样，其他测试的初始偏航角从 $U(-\pi/6, \pi/6)$ 中采样。足部与地面之间的摩擦系数从 $U(0.4, 1.0)$ 中采样。外力在横向方向上施加5秒。(B-D) TCN- N 编码器中记忆长度 N 的重要性。(E-G) 特权训练的重要性。(F) 教师（灰色）和直接训练、未经特权训练的TCN-20学生（红色）的学习曲线。作为对比，蓝色线表示经过特权训练的TCN-20学生的平均奖励。该奖励是通过在均匀采样地形上运行每个策略计算得出的。(H-J) 自适应课程的重要性。

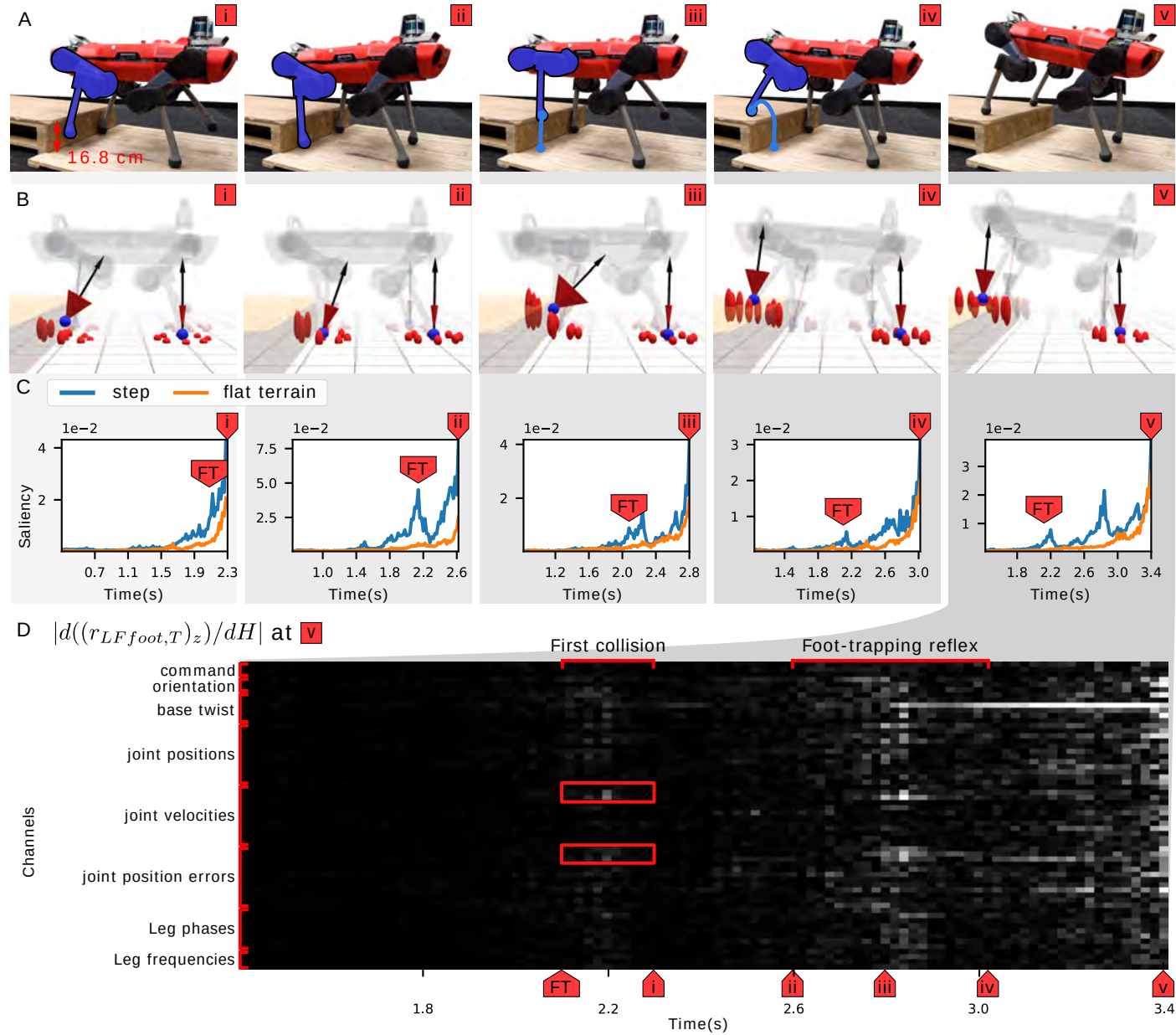


Fig. 6. Analysis of the emergent foot-trapping reflex. FT denotes the first contact of the LF foot with the step (foot-trapping event). (A) The LF foot hits the step and then manifests higher foot clearance to overcome the step (ii-iv) in the following swing phase. (B) Reconstructed terrain information from TCN embeddings. Red ellipsoid: estimated terrain shape around the foot. The center of the ellipsoid refers to the estimated terrain elevation and the vertical length represents uncertainty (standard deviation). Black arrow: terrain normal at the in-contact foot. Red cone: uncertainty of normal estimation. Blue spheres: estimated in-contact feet. (C) Input saliency at different moments. The peaks show that the TCN policy attends to the foot-trapping (FT) that happened around 2.1 s. The orange curve (flat terrain) shows the saliency value computed on a flat terrain at similar gait phases. (D) Saliency map unrolled across input channels at 3.4 s. Red boxes refer to joint measurements from the LF leg at the moment it collides with the step.

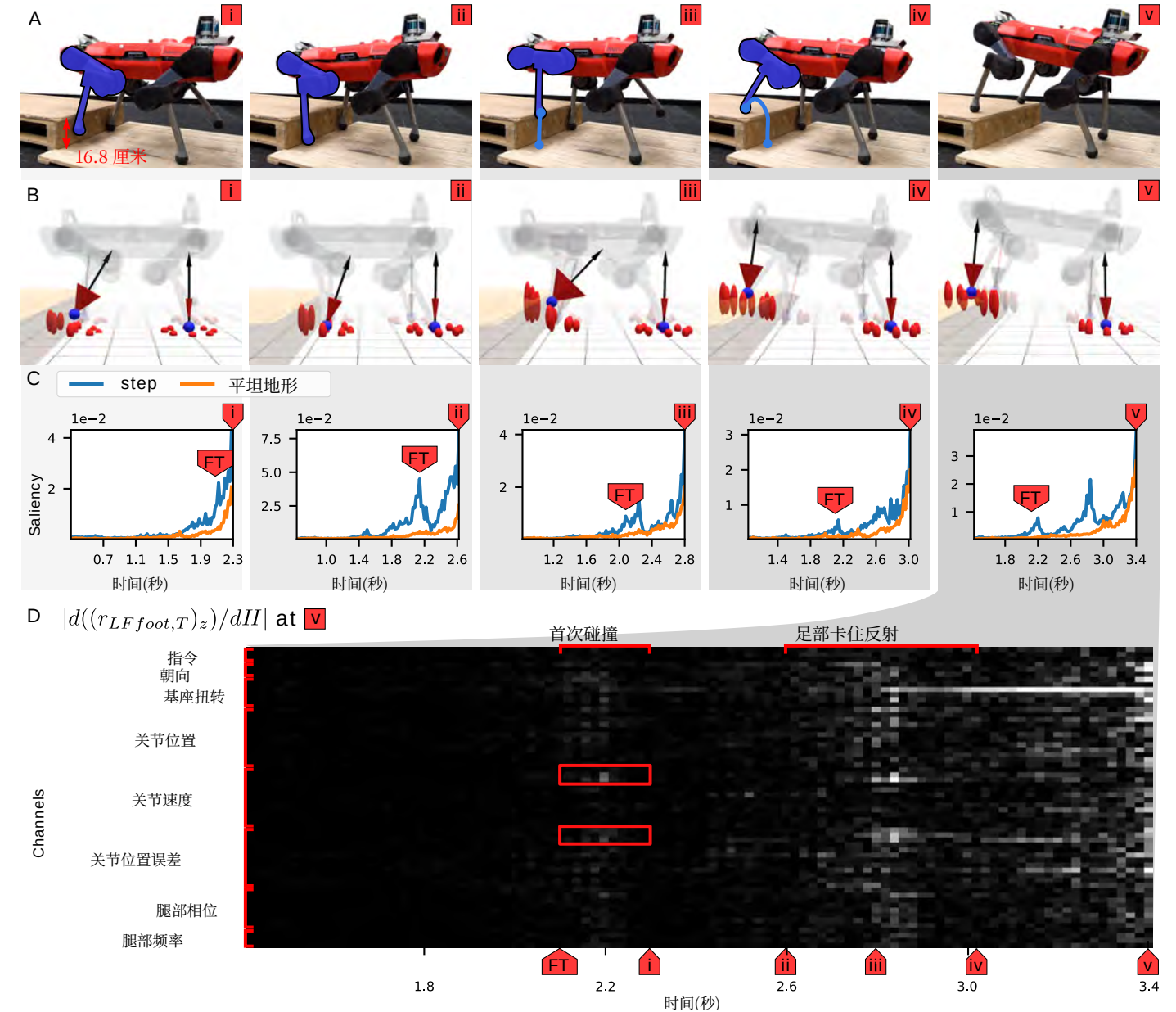


图 6. 涌现的足部卡住反射分析。 FT 表示左前足首次接触台阶（足部绊倒事件）。(A) 左前足撞击台阶后，在随后的摆动相中表现出更高的足部离地高度以越过台阶（ii-iv）。(B) 从 TCN 嵌入重建的地形信息。红色椭球体：足部周围估计的地形形状。椭球体中心表示估计的地形高程，垂直长度代表不确定性（标准差）。黑色箭头：接触足处的地形法线。红色锥体：法线估计的不确定性。蓝色球体：估计的接触足。(C) 不同时刻的输入显著性。峰值显示 TCN 策略关注发生在约 2.1 秒的足部绊倒事件 (FT)。橙色曲线（平坦地形）显示了在相似步态相位下平坦地形上计算的显著性值。(D) 3.4 秒时跨输入通道展开的显著性图。红色框表示左前腿在与台阶碰撞时刻的关节测量值。

Validation of the method

We present ablation studies to justify each component of our approach: (1) using a sequence model for the student policy, (2) privileged training, and (3) adaptive terrain curriculum.

Memory in proprioceptive control

We evaluate the importance of incorporating proprioceptive memory in the controller via the TCN architecture [22]. Let TCN- N denote a TCN with a receptive field of N time steps. The network architectures we use are specified in detail in Table S5. We test controllers in diagnostic settings designed to focus on specific capabilities. Specifically, we test omnidirectional locomotion on sloped ground, traversal of a discrete step, and robustness to external disturbances (Fig. 5A).

Fig. 5B-D summarizes the importance of the memory length N . In these experiments, N is varied from 1 (corresponding to 20 ms of memory) to 100 (2 s of proprioceptive memory). The latter is the default setting used in our deployed controller.

As shown in Fig. 5B, memory length doesn't have a strong effect in the uniform slope setting. Memory length does have a strong effect on the controller's ability to traverse a step (Fig. 5B-C). Controllers with longer memory are able to handle higher steps. As shown in Fig. 5C, the failure rate of limited-memory controllers is particularly high when the hind legs encounter the step. Controllers with longer memory also adapt hind leg trajectories to ensure higher foot clearances.

Fig. 5D shows that controllers with longer memory are more robust to external disturbances. We applied an external 50 N force laterally to the base for 5 s during a straight walk and evaluated the resulting deviation from the intended locomotion direction. The deviation of the TCN-100 controller was 35.5 % lower than that of TCN-1.

Privileged training

We now assess the importance of privileged training. As a baseline, we train a TCN-20 policy directly, without the two-stage privileged training protocol. The policy is trained by TRPO [36] with the same reward and hyperparameters that we use for teacher training. This baseline is compared to the same TCN-20 architecture trained via privileged learning.

The results are summarized in Fig. 5E-G. Fig. 5E shows that the baseline fails the diagnostic tests: it is incapable of locomoting on a slope or traversing a step. Fig. 5F shows that the baseline does not reach comparable reward during training as the teacher MLP architecture with privileged information or the proprioceptive TCN-20 architecture (same as the baseline, no privileged information) trained via privileged learning. Fig. 5G shows the mean episode length during training, which indicates that the baseline fails to learn to balance and locomote.

Adaptive terrain curriculum

We now evaluate the effect of the adaptive terrain curriculum on teacher training. Terrains used for training (specifically, hills, steps, and stairs) are shown in Fig. 4B. As a baseline, we trained a teacher using randomly generated terrains that are uniformly sampled from $\mathcal{C}$ as specified in Table S2. The success rates on the testing terrains are significantly lower when trained without the adaptive curriculum, as shown in Fig. 5H. Fig. 5I shows that a teacher trained without adaptive curriculum plateaus at a lower reward level. Throughout the training process, the mean episode length is shorter for the model being trained without adaptive curriculum (Fig. 5J). This is because uniform sampling is more likely to draw terrains that cannot be successfully traversed by the policy being trained. On these terrains, the policy fails early and receives less training signal as a result. The adaptive curriculum modulates the difficulty of sampled terrains so as to maximize the didactic benefit of each episode. We provide an additional evaluation of the adaptive curriculum in supplementary section S6.

Further analysis of emergent behavior

Here we provide further analysis on how the proprioceptive policy adapts to different situations.

To investigate how the proprioceptive policy perceives the environment, we trained a decoder network which reconstructs the privileged information $x_t \in X$ from the output of an intermediate layer of a trained TCN policy. x_t consists of information that is not directly observable by the student policy such as contact states, terrain shape, and external disturbances. For classification of foot contact states, we employ a standard cross-entropy loss function. For regression of other states, we predict both mean m_i and standard deviation σ_i for each component and use a negative Gaussian log-likelihood loss to quantify the uncertainty encoded in the TCN representation [45]:

$$\mathcal{L} = \sum_{i \in \dim(X \setminus \text{contact states})} \frac{(m_i - m_i^{gt})^2}{2\sigma_i^2} + \log(\sigma_i) \quad (8)$$

with added weight decay. The superscript gt refers to the ground truth generated in simulation. Note that the parameters of the policy network are fixed during decoder training. Therefore, the decoder network is not used for policy training. It only provides insight into the information encoded by the TCN policy after training.

In Fig. 6, we provide snapshots of the foot-trapping reflex motion (Fig. 6A) and the reconstructed privileged information. In Fig. 6B we show the reconstructed terrain geometry and foot contact state. When the LF foot collides with the step, the estimated elevation in front of the front legs increases and its uncertainty grows (i+ii). The estimated elevations and normal vectors adapt to the step during the foot-trapping reflex (iii+iv). After the successful step-up, the terrain uncertainty remains elevated (v), indicating an anticipation of generally rough terrain. Additionally, the decoder network can detect foot contacts with horizontal and vertical surfaces while successfully identifying frontal collision as such, as indicated by the estimated terrain normal vector (i + iii). The ability to reconstruct explicit environmental information from the encoding of the proprioceptive history is a strong indicator that the TCN policy learns to build an internal representation of the environment and uses it for decision making. We provide more examples of the reconstructed privileged information in supplementary section S7.

We then analyse how the proprioceptive policy leverages past observations. We compute the saliency map of the input $H \in \mathbb{R}^{60 \times N}$, and visualize the sensitivity of the policy to each element of the input while overcoming the step [46]. Each column of H is a proprioceptive measurement $h \in \mathbb{R}^{60}$, and we stack N measurements (history length = $0.02 \text{ s} \times N$). We define the saliency value for the i -th measurement ($i \in [0, N]$) as

$$M_i = \sum_{j \in \text{channels}} (|d((r_{f,T})_z)/dH_{i,j}|) \in \mathbb{R}, \quad (9)$$

where $(r_{f,T})_z$ refers to the height command for the foot f . We computed the value for $(r_{f,T})_z$ because we are interested in the change in foot clearance. M_i can be interpreted as the sensitivity of the output to the i th measurement. As we use 1D convolution over time, the output is in $\mathbb{R}^N$, i.e., each row of H is regarded as a channel.

In Fig. 6C we can see that the saliency value at the foot-trapping event (FT) is kept high while stepping up. The policy has direct access to the measurements at the moment of foot-trapping, and leverages this in the following swing phase. This is highlighted by the red boxes in Fig. 6D. The policy attends to the LF leg joint states measured at the foot-trapping event.

方法验证

我们通过消融研究来论证方法中每个组件的必要性: (1) 为学生策略使用序列模型, (2) 特权信息训练, 以及 (3) 自适应地形课程。

本体感觉控制中的记忆

我们通过时序卷积网络架构评估了在控制器中融入本体感觉记忆的重要性 [22]。设 TCN- N 表示感受野为 N 个时间步的时序卷积网络。我们使用的网络架构在表S5中有详细说明。我们在专门针对特定能力设计的诊断设置中测试控制器。具体来说, 我们测试了斜坡地面上的全向运动、离散台阶的跨越以及对外部干扰的鲁棒性(图5A)。

图5B-D总结了记忆长度 N 的重要性。在这些实验中, N 从 1 (对应20毫秒的记忆) 变化到100 (2秒的本体感觉记忆)。后者是我们部署的控制器中使用的默认设置。

如图5B所示, 在均匀斜坡设置中, 记忆长度没有显著影响。但记忆长度对控制器跨越台阶的能力有显著影响(图5B-C)。记忆长度更长的控制器能够处理更高的台阶。如图5C所示, 当后腿遇到台阶时, 有限记忆控制器的失败率特别高。记忆长度更长的控制器还会调整后腿轨迹, 以确保更高的足部离地高度。

图5D显示, 记忆长度更长的控制器对外部干扰具有更强的鲁棒性。我们在直线行走过程中, 对基座横向施加了持续5秒的外部50N力, 并评估了由此产生的运动方向偏差。TCN-100控制器的偏差比TCN-1低了35.5%。

特权训练

我们现在评估特权训练的重要性。作为基线, 我们直接训练一个TCN-20策略, 而不采用两阶段特权训练协议。该策略通过 TRPO [36] 进行训练, 使用与教师训练相同的奖励和超参数。将此基线与通过特权学习训练的相同TCN-20架构进行比较。

结果总结于图5E-G。图5E显示基线未能通过诊断测试: 它无法在斜坡上移动或跨越台阶。图5F显示, 在训练期间, 基线未能达到与拥有特权信息的教师MLP架构或通过特权学习训练的本体感觉TCN-20架构(与基线相同, 无特权信息)相当的奖励。图5G显示了训练期间的平均回合长度, 这表明基线未能学会平衡和移动。

自适应地形课程

我们现在评估自适应地形课程对教师训练的影响。用于训练的地形(具体包括丘陵、台阶和楼梯)如图4B所示。作为基线, 我们使用从 $\mathcal{C}$ 中均匀采样的随机生成地形训练了一个教师, 具体参数见表S2。在没有自适应课程的情况下训练时, 测试地形上的成功率显著降低, 如图5H所示。图5I显示, 未使用自适应课程训练的 教师奖励水平较早进入平台期。在整个训练过程中, 未使用自适应课程训练的模型平均回合长度更短(图5J)。这是因为均匀采样更倾向于抽取当前训练策略无法成功穿越的地形。在这些地形上, 策略会提前失败, 从而获得的训练信号也更少。自适应课程通过调节采样地形的难度, 最大化每个回合的教学收益。我们在补充章节S6中提供了对自适应课程的额外评估。

涌现行为的进一步分析

在此, 我们进一步分析本体感受策略如何适应不同情境。

为了研究本体感受策略如何感知环境, 我们训练了一个解码器网络, 该网络从训练好的TCN策略的中间层输出中重建特权信息 $x_t \in X$ 。 x_t 包含学生策略无法直接观测到的信息, 例如接触状态、地形形状和外部干扰。对于足部接触状态的分类, 我们采用标准的交叉熵损失函数。对于其他状态的回归, 我们预测每个分量的均值 m_i 和标准差 σ_i , 并使用负高斯对数似然损失来量化TCN表征中编码的不确定性 [45]:

$$\mathcal{L} = \sum_{i \in \dim(X \setminus \text{contact states})} \frac{(m_i - m_i^{gt})^2}{2\sigma_i^2} + \log(\sigma_i) \quad (8)$$

并加入了权重衰减。上标 gt 指的是仿真中生成的真实值。请注意, 在解码器训练期间, 策略网络的参数是固定的。因此, 解码器网络不用于策略训练。它仅用于洞察训练后TCN策略所编码的信息。

在图6中, 我们展示了足部卡住反射运动(图6A)和重建的特权信息的快照。在图6B中, 我们展示了重建的地形几何和足部接触状态。当左前足与台阶碰撞时, 前腿前方的估计高程增加, 其不确定性也随之增长 (i+ii)。在足部卡住反射过程中, 估计高程和法向量适应了台阶 (iii+iv)。成功上台阶后, 地形不确定性仍然较高 (v), 表明对一般崎岖地形的预期。此外, 解码器网络能够检测与水平和垂直表面的足部接触, 同时成功识别正面碰撞, 如估计的地形法向量所示 (i + iii)。从本体感受历史的编码中重建显式环境信息的能力, 有力地表明TCN策略学会了构建环境的内部表征, 并将其用于决策。我们在补充章节S7中提供了更多重建特权信息的示例。

我们随后分析了本体感受策略如何利用过去的观测信息。我们计算了输入 $H \in \mathbb{R}^{60 \times N}$ 的显著性图, 并可视觉化了策略在克服台阶 [46] 时对输入中每个元素的敏感度。 H 的每一列都是一个本体感受测量值 $h \in \mathbb{R}^{60}$, 我们堆叠了 N 个测量值 (历史长度 = $0.02 \text{ 秒} \times N$)。我们将第 i 个测量值 ($i \in [0, N]$) 的显著性值定义为

$$M_i = \sum_{j \in \text{channels}} (|d((r_{f,T})_z)/dH_{i,j}|) \in \mathbb{R}, \quad (9)$$

其中 $(r_{f,T})_z$ 指的是足 f 的高度指令。我们计算了 $(r_{f,T})_z$ 的值, 因为我们关注的是足部离地高度的变化。 M_i 可以解释为输出对第 i 个测量值的敏感度。由于我们使用一维卷积处理时间维度, 输出为 $\mathbb{R}^N$, 即 H 的每一行都被视为一个通道。

在图6C中, 我们可以看到, 在迈上台阶时, 足部绊倒事件 (FT) 处的显著性值保持较高水平。策略可以直接获取足部绊倒时刻的测量值, 并在随后的摆动相中加以利用。这一点由图 6D中的红色方框突出显示。策略关注的是在足部绊倒事件中测量到的左前腿关节状态。

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SUPPLEMENTARY MATERIALS

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https://github.com/leggedrobotics/learning_四足机器人_运动_在_具有挑战性的_地形_补充。

补充材料

第S1节 术语表 第S2节 实现细节 第S3节 足部轨迹生成器 第S4节 教师策略训练的奖励函数 第 S5 节 参数化地形 第 S6 节 自适应地形课程的定性评估 第S7节 不同情境下特权信息的重建 第S8节 循环神经网络学生策略 第 S9 节 学生训练中潜在表示损失的消融实验 算法S1 基于自动地形课程的教师训练 图S1 自适应课程示意图。 图S2 不同情境下重建的特权信息。 图S3 本体感受控制器的神经网络架构比较 表S1 训练的计算时间 表S2 模拟地形的参数空间 $\mathcal{C}$ 表S3 自动地形课程的超参数 表S4 本体感受控制器的状态表示与特权信息 表S5 神经网络架构 表S6 学生策略的网络参数设置与训练时间 表S7 教师策略训练的超参数 表S8 学生策略训练的超参数 表S9 解码器训练的超参数**视频S1 森林部署 视频 S2 不稳定碎片上的运动 视频 S3 台阶实验 视频 S4 负载实验 视频S5 足端打滑实验**

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SUPPLEMENTARY MATERIALS

S1. Nomenclature

$\hat{(\cdot)}$	normalized vector
$\dot{(\cdot)}$	first derivative
$\ddot{(\cdot)}$	teacher’s quantity
$\dot{(\cdot)}_T$	target quantity
${}^C_{AB}\boldsymbol{v}$	linear velocity of B frame with respect to A frame expressed in C frame
$\boldsymbol{c}_T$	terrain parameter vector
$\boldsymbol{\omega}$	angular velocity
$\boldsymbol{\tau}$	joint torque
θ	joint angle
ψ	yaw angle
ϕ	leg phase
f	leg frequency
$\boldsymbol{r}_f$	linear position of a foot
$\boldsymbol{e}_g$	gravity vector
H	horizontal frame
g_i	gap function of the i -th possible contact pair
I_c	index set of all contacts
$I_{c,body}$	index set of body contacts
$I_{c,foot}$	index set of foot contacts
I_{swing}	index set of swing legs
$ \cdot $	cardinality of a set or l_1 norm
$\ \cdot\ $	l_2 norm

S2. Implementation details

The RaiSim simulator [30] is used for rigid-body and contact dynamics simulation. The actuator networks [12] are trained for each robot to simulate Series Elastic Actuators (SEA) [47] at the joint. The input to the actuator model is a 6-dimensional real-valued vector consisting of the joint position error and velocity at current time step t and two past states corresponding to $t - 0.01$ s and $t - 0.02$ s. The feature selection is done as in [12].

As several studies have shown that randomization of dynamic properties improves the robustness of the policy [12, 16], we also randomized several physical quantities, and the teacher policy has access to these values during training. We applied disturbances, randomized friction coefficients between the feet and the terrain, and additive noise to the observations during training.

The training process for the teacher policy is depicted in Algorithm S1. Hyperparameters are given in Table S3. In our implementation of the terrain curriculum, we update the curriculum every $N_{evaluate}$ policy iterations to reduce variance. We assume that within $N_{evaluate}$ iterations, the performance of the policy is similar. With the slower update rate, the measurement probability of Eq.6 becomes

$$\Pr(y_j^k | c_{T,j}^k) \approx \sum_{N_{evaluate}} \sum_{N_{traj}} \frac{\mathbb{1}(\text{Tr}(c_{T,j}^k, \pi) \in [0.5, 0.9])}{N_{traj} N_{evaluate}}. \quad (10)$$

Additionally, we leverage replay memory to prevent degeneration of the particle filter and to avoid catastrophic forgetting.

The controller is implemented with a state machine to switch between the “standing still” state and the locomotion state. We set the base frequency f_0 to zero when the zero command is given for 0.5 s, which stops FTGs, and the robot stands still on the terrain. f_0 is set to 1.25 Hz when the direction command is given or the linear velocity of the base exceeds 0.3 m/s for the disturbance rejection. The state machine is included in the training environment.

During the deployment, the base velocity and orientation are estimated by the state estimator that relies on inertial measurements and leg kinematics [37].

The neural network policy runs at 400 Hz on an onboard CPU (Intel i7-5600U, 2.6 – 3.2GHz, dual-core 64-bit) integrated into the robot. The Tensorflow C++ API is used for onboard inference.

Algorithm S1. Teacher training with automatic terrain curriculum

```

1: Initialize a replay memory, Sample  $N_{particle}$   $\boldsymbol{c}_{T,0}$ s uniformly from  $\mathcal{C}$  (Table S2),  $i, j = 0$ .
2: repeat
3:   for  $0 \leq k \leq N_{evaluate}$  do
4:     for  $0 \leq l \leq N_{particle}$  do
5:       for  $0 \leq m \leq N_{traj}$  do
6:         Generate terrain using  $\boldsymbol{c}_{T,j}^l$ 
7:         Initialize robot at random position
8:         Run policy  $\pi_i$ 
9:         Compute traversability label for each state transition (Eq. 2)
10:        Save the scores and the trajectory
11:        Update policy using TRPO [36]
12:         $i = i + 1$ 
13:      for  $0 \leq l \leq N_{particle}$  do
14:        Compute measurement probability for each parameter  $\boldsymbol{c}_{T,j}^l$  (Eq. 9)
15:      for  $0 \leq l \leq N_{particle}$  do
16:        Update weights  $w_j = \frac{P(y_i^l | c_{T,j}^l)}{\sum_m P(y_i^m | c_{T,j}^m)}$ 
17:      Resample  $N_{particle}$  parameters
18:      Append  $\boldsymbol{c}_{T,j}$ s to the replay memory
19:      for  $0 \leq l \leq N_{particle}$  do
20:        by  $p_{replay}$  probability, sample from replay memory
21:        by  $p_{transition}$  probability, move  $\boldsymbol{c}_{T,j}^l$  to an adjacent value in  $\mathcal{C}$ .
22:       $j = j + 1$ 
23: until Convergence

```

S3. Foot trajectory generator

The foot trajectory is defined as

$$F(\phi_i) = \begin{cases} (h(-2k^3 + 3k^2) - 0.5)^{H_i} z & k \in [0, 1] \\ (h(2k^3 - 9k^2 + 12k - 4) - 0.5)^{H_i} z & k \in [1, 2] \\ -0.5^{H_i} z & \text{otherwise,} \end{cases} \quad (11)$$

where $k = 2(\phi_i - \pi)/\pi$ and h is a parameter for the maximum foot height. Each segment during the swing phase ($k \in [0, 2)$) is a cubic Hermite spline connecting the highest and lowest points with a zero first derivative at the connecting points. Other periodic functions such as $h_i \sin(\phi_i)$ can be used for the FTG. With a set of reasonably tuned f_0 , h and $\phi_{i,0}$, a quadruped can stably step in place. In our setting, $f_0 = 1.25$, $h = 0.2$ m, and $\phi_{i,0}$ are sampled from $U(0, 2\pi)$.

S4. Reward function for teacher policy training

The reward function is defined as $0.05r_{lv} + 0.05r_{av} + 0.04r_b + 0.01r_{fc} + 0.02r_{bc} + 0.025r_s + 2 \cdot 10^{-5}r_\tau$. The individual terms are defined as follows.

- Linear Velocity Reward (r_{lv}): This term maximizes the $v_{pr} = (\hat{B}_{IB}^B v)_{xy} \cdot (\hat{B}_{IB}^B \hat{v}_T)_{xy}$, which is the base linear velocity projected onto the command direction.

$$r_{lv} := \begin{cases} \exp(-2.0(v_{pr} - 0.6)^2) & v_{pr} < 0.6 \\ 1.0 & v_{pr} \geq 0.6 \\ 0.0 & \text{zero command} \end{cases}. \quad (12)$$

补充材料

S1. 术语表

$\hat{(\cdot)}$ 归一化向量 $\dot{(\cdot)}$ 一阶导数 $\ddot{(\cdot)}$ 教师量 $\dot{(\cdot)}_T$ 目标量 CABv B框架相对于A框架的线速度 $\dot{(\cdot)}$ 在C框架中表示 $\boldsymbol{c}_T$ 地形参数向量 $\boldsymbol{\omega}$ 角速度 $\boldsymbol{\tau}$ 关节扭矩 θ 关节角度 ψ 偏航角 ϕ 腿部相位 f 腿部频率 $\boldsymbol{r}_f$ 足的线位置 $\boldsymbol{e}_g$ 重力向量 H 水平框架 g_i 第i个可能接触对的间隙函数 I_c 所有接触的索引集 $I_{c,body}$ 身体接触的索引集 $I_{c,foot}$ 足部接触的索引集 I_{swing} 摆动腿的索引集 $|\cdot|$ 集合的基数或 l_1 范数 $\|\cdot\|$ l_2 范数

S2. 实现细节

RaiSim模拟器 [30] 用于刚体与接触动力学仿真。执行器网络 [12] 针对每个机器人进行训练，以模拟关节处的串联弹性致动器（SEA） [47]。执行器模型的输入是一个6维实值向量，包含当前时间步 t 的关节位置误差和速度，以及对应 $t - 0.01$ 秒和 $t - 0.02$ 秒的两个过去状态。特征选择按照 [12]中的方法进行。

多项研究表明，对动态属性进行随机化能够提升策略的鲁棒性 [12, 16]，，因此我们也对若干物理量进行了随机化处理，并且教师策略在训练过程中可以获取这些值。我们在训练中施加了扰动，对足部与地形之间的摩擦系数进行了随机化，并在观测值中添加了加性噪声。

教师策略的训练过程如算法S1所示。超参数列于表S3中。在我们的地形课程实现中，我们每 $N_{evaluate}$ 次策略迭代更新一次课程，以降低方差。我们假设在 $N_{evaluate}$ 次迭代内，策略的性能是相似的。随着更新速率的降低，公式(6)的测量概率变为

$$\Pr(y_j^k | c_{T,j}^k) \approx \sum_{N_{evaluate}} \sum_{N_{traj}} \frac{\mathbb{1}(\text{Tr}(c_{T,j}^k, \pi) \in [0.5, 0.9])}{N_{traj} N_{evaluate}}. \quad (10)$$

此外，我们利用回放记忆来防止粒子滤波器的退化，并避免灾难性遗忘。

控制器通过状态机实现，在“静止站立”状态与运动状态之间切换。当零指令持续0.5秒时，我们将基频 f_0 设为零，从而停止FTGs，机器人在地形上静止站立。当给出方向指令或为进行扰动抑制而使得基座线速度超过0.3米/秒时， f_0 被设置为1.25赫兹。该状态机包含在训练环境中。

在部署过程中，基础速度和朝向由状态估计器估算，该估计器依赖于惯性测量和腿部运动学 [37]。

神经网络策略以400赫兹的频率在集成于机器人的板载CPU（Intel i7-5600U，2.6–3.2GHz，双核64位）上运行。TensorFlow C++ API用于板载推理。

算法S1. 基于自动地形课程的教师训练

```

1: 初始化一个回放记忆，从(表S2)中均匀采样 $N_{particle}$   $\boldsymbol{c}_{T,0}$ 
    $\mathcal{C}$  (表S2),  $i, j = 0$ .
2: 重复
3: 对于  $0 \leq k \leq N_{evaluate}$  执行
4:   对于  $0 \leq l \leq N_{particle}$  执行
5:     对于  $0 \leq m \leq N_{traj}$  执行
6:       使用 $\boldsymbol{c}_{T,j}^l$ 生成地形
7:       在随机位置初始化机器人
8:       运行策略  $\pi_i$ 
9:       计算每个状态的可遍历性标签
         状态转移（公式2）
10:      保存分数和轨迹
11:      使用 TRPO [36] 更新策略
12:       $i = i + 1$ 
13:    for  $0 \leq l \leq N_{particle}$  do
14:      计算每个的测量概率
        参数  $\boldsymbol{c}_{T,j}$ s （公式 9）
15:    for  $0 \leq l \leq N_{particle}$  do
16:      更新权重  $w_j = \frac{P(y_i^l | c_{T,j}^l)}{\sum_m P(y_i^m | c_{T,j}^m)}$ 
17:    重采样 $N_{particle}$  参数
18:    将 $\boldsymbol{c}_{T,j}$ s 追加到回放记忆
19:    for  $0 \leq l \leq N_{particle}$  do
20:      以  $p_{replay}$  概率，从回放记忆中采样
21:      以  $p_{transition}$  概率，将 $\boldsymbol{c}_{T,j}$  移动至相邻值
        在  $\mathcal{C}$  中。
22:     $j = j + 1$ 
23: 直到收敛

```

S3. 足部轨迹生成器

足部轨迹定义为

$$F(\phi_i) = \begin{cases} (h(-2k^3 + 3k^2) - 0.5)^{H_i} z & k \in [0, 1] \\ (h(2k^3 - 9k^2 + 12k - 4) - 0.5)^{H_i} z & k \in [1, 2] \\ -0.5^{H_i} z & \text{otherwise,} \end{cases} \quad (11)$$

其中 $k = 2(\phi_i - \pi)/\pi$ 且 h 是最大足高的参数。摆动相 ($k \in [0, 2)$) 中的每个分段均为三次埃尔米特样条，连接最高点和最低点，且在连接点处一阶导数为零。其他周期函数如 $h_i \sin(\phi_i)$ 也可用于FTG。通过一组合理调优的 $f = 1.25$, $= 0.2$ $\phi_{i,0}$ 0.2π h 和 ϕ_i ，四

足机器人能够稳定地原地踏步。在我们的设定中， f, h . m 和从 $U(\cdot)$ 中采样。

S4. 教师策略训练的奖励函数

奖励函数定义为 $0.05r_{lv} + 0.05r_{av} + 0.04r_b + 0.01r_{fc} + 0.02r_{bc} + 0.025r_s + 2 \cdot 10^{-5}r_\tau$ 。各项具体定义如下。

- 线速度奖励 (r_{lv})：该项最大化 $v_{pr} = (\hat{B}_{IB}^B v)_{xy} \cdot (\hat{B}_{IB}^B \hat{v}_T)_{xy}$ ，即投影到指令方向上的基座线速度。

$$r_{lv} := \begin{cases} \exp(-2.0(v_{pr} - 0.6)^2) & v_{pr} < 0.6 \\ 1.0 & v_{pr} \geq 0.6 \\ 0.0 & \text{zero command} \end{cases}. \quad (12)$$

The velocity threshold is defined as 0.6 m/s which is the maximum speed reachable on the flat terrain with the existing controller [26].

- Angular Velocity Reward (r_{av}): We motivate the agent to turn as fast as possible along the base z-axis when $(\hat{B}_{IB}\hat{\omega}_T)_z$ is nonzero. It is defined as

$$r_{av} := \begin{cases} \exp(-1.5(\omega_{pr} - 0.6)^2) & \omega_{pr} < 0.6 \\ 1.0 & \omega_{pr} \geq 0.6 \end{cases}, \quad (13)$$

where $\omega_{pr} = (\hat{B}_{IB}\omega)_z \cdot (\hat{B}_{IB}\hat{\omega}_T)_z$.

- Base Motion Reward (r_b): This term penalizes the velocity orthogonal to the target direction and the roll and pitch rates such that the base is stable during the locomotion.

$$r_b := \exp(-1.5v_o^2) + \exp(-1.5|(\hat{B}_{IB}\omega)_{xy}|^2) \quad (14)$$

where $v_o = |(\hat{B}_{IB}v)_{xy} - v_{pr} \cdot (\hat{B}_{IB}\hat{v}_T)_{xy}|$. When the stop command is given, v_o is replaced by $|(\hat{B}_{IB}v)|$.

- Foot Clearance Reward (r_{fc}): When a leg is in swing phase, i.e., $\phi_i \in [\pi, 2\pi)$, the robot should lift the corresponding foot higher than the surroundings to avoid collision. We first define the set of such collision-free feet as $\mathcal{F}_{clear} = \{i : r_{f,i} > \max(H_{scan,i}), i \in I_{swing}\}$, where $H_{scan,i}$ is the set of scanned heights around the i -th foot. Then the clearance cost is defined as

$$r_{fc} := \sum_{i \in I_{swing}} (\mathbb{1}_{\mathcal{F}_{clear}}(i) / |I_{swing}|) \in [0.0, 1.0]. \quad (15)$$

- Body Collision Reward (r_{bc}): We want to penalize undesirable contact between the robot's body parts and the terrain to avoid hardware damage.

$$r_{bc} := -|I_{c,body} \setminus I_{c,foot}|. \quad (16)$$

- Target Smoothness Reward (r_s): The magnitude of the second order finite difference derivatives of the target foot positions are penalized such that the generated foot trajectories become smoother.

$$r_s := -|(r_{f,d})_t - 2(r_{f,d})_{t-1} + (r_{f,d})_{t-2}|. \quad (17)$$

- Torque Reward (r_τ): We penalize the joint torques to prevent damaging joint actuators during the deployment and to reduce energy consumption ($\tau \propto$ electric current).

$$r_\tau := -\sum_{i \in joints} |\tau_i|. \quad (18)$$

S5. Parameterized terrains

It is important to generate training environments that can pose representative challenges such as foot slippage and foot-trapping. To efficiently synthesize random terrains, we use procedural generation techniques [48]. This method allows us to generate a large number of different terrains by changing a set of terrain parameters $c_T \in \mathcal{C}$. In the following, we describe the three terrain generators used in this work. See Fig. 4B for a visualization of the terrains and Table S2 for the definition of parameter spaces $\mathcal{C}$.

- The *Hills* terrain is based on Perlin noise [49]. The terrain is generated via three parameters: roughness, frequency of the Perlin noise, and amplitude of the Perlin noise. The height of each element of the output height map hm is defined as $hm[i, j] := \text{Perlin}(c_{T,2}, c_{T,3})[i, j] + U(-c_{T,1}, c_{T,1})$. A policy experiences smooth slopes and foot slippage on this terrain during training.

- The *Steps* terrain consists of square steps of random height. For every $c_{T,1}$ by $c_{T,1}$ blocks, the height is sampled from $U(0, c_{T,2})$. A policy experiences discrete elevation changes and foot-trapping on this terrain.

- The *Stairs* terrain is a staircase with fixed width and height. The robot is initialized at the flat segment in the middle of the staircase (see Fig. 4B).

The ranges are defined considering the kinematics of the robot, e.g., a step height should be lower than leg length. During training, the terrain is regenerated every episode with a different random seed.

S6. Qualitative evaluation of the adaptive terrain curriculum

The behavior of adaptive curriculum is illustrated in Fig. S1. Fig. S1A-C focuses on the *Hills* terrain type. There are three parameters for this terrain: roughness, frequency, and amplitude. The relationship between traversability (Eq. 3) and desirability (Eq. 4) is illustrated in Fig. S1A-B. Undesirable terrains are either too easy or too difficult, as shown in the leftmost and rightmost panels of Fig. S1A. Fig. S1B-C shows that the particle filter fits the latent distribution of desirable terrains, which has a bow shape in the frequency-amplitude marginal (middle). Fig. S1D focuses on the *Stairs* terrain type and shows the evolution of terrain parameters during training. The particle filter rejects parameters that represent short and steep steps (upper-left area). The curriculum initially focuses on wide and shallow steps (middle panels, particularly Iter. 50-60), and then broadens the distribution to include narrower steps (rightmost panels).

S7. Reconstruction of the privileged information in different situations

In Fig. S2, we provide the decoded privileged information in different situations. Fig. S2A shows the estimated friction coefficient between the feet and the terrain when traversing a wet, slippery whiteboard, as shown in [Movie S5](#). The estimate decreases as soon as the first foot starts slipping (i), remains low throughout the traversal (ii) and increases about 2 s after the robot returns to normal ground (iii). The external disturbance and terrain information can also be reconstructed from the TCN embedding. As shown in Fig. S2B, the decoder detects downward external force when an unknown 10 kg payload is applied. While traversing dense vegetation as shown in Fig. S2C, it detects a force opposite the motion direction, which makes the policy to counter-act and push through the vegetation. The uncertainty of the elevation estimates are notably high in the natural terrains shown in Fig. S2C and Fig. S2D, which indicates that the TCN policy encodes the roughness of the terrain.

S8. Recurrent neural network student policy

We use the TCN architecture for the proprioceptive policy [22]. For comparison, we also evaluated a recurrent network with gated recurrent units (GRU) [50]. The architectures are specified in Tables S5 and S6. The loss function for training a GRU student policy is defined as

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t))^2 + (\bar{l}_t(o_t, x_t) - l_t(o_t))^2. \quad (19)$$

To improve the performance and computational efficiency of the training, we have implemented Truncated Backpropagation Through Time (Truncated BPTT) [51].

Performance on the diagnostic settings presented in Fig. 5A is given in Fig. S3. Overall, the performance of the GRU-based controller is between that of TCN-20 and TCN-100. The performance is comparable to TCN-100 in the slope setting, but the GRU-based controller fails to achieve the performance of TCN-100 in step experiments.

速度阈值定义为0.6米/秒，这是在现有控制器下在平坦地形上可达到的最大速度 [26]。

- 角速度奖励 (r_{av}): 当沿基座 z 轴的 $(\hat{B}_{IB}\hat{\omega}_T)_z$ 非零时，我们激励智能体尽可能快地转动。其定义为

$$r_{av} := \begin{cases} \exp(-1.5(\omega_{pr} - 0.6)^2) & \omega_{pr} < 0.6 \\ 1.0 & \omega_{pr} \geq 0.6 \end{cases}, \quad (13)$$

其中 $\omega_{pr} = (\hat{B}_{IB}\omega)_z \cdot (\hat{B}_{IB}\hat{\omega}_T)_z$.

- 基座运动奖励 (r_b): 该项惩罚与目标方向正交的速度以及横滚和俯仰角速率，以确保基座在运动过程中保持稳定。

$$r_b := \exp(-1.5v_o^2) + \exp(-1.5|(\hat{B}_{IB}\omega)_{xy}|^2) \quad (14)$$

其中 $v_o = |(\hat{B}_{IB}v)_{xy} - v_{pr} \cdot (\hat{B}_{IB}\hat{v}_T)_{xy}|$ 。当给出停止指令时， v_o 被替换为 $|(\hat{B}_{IB}v)|$ 。

- 足部离地奖励 (r_{fc}): 当腿部处于摆动相时，即 $\phi_i \in [\pi, 2\pi)$ ，机器人应将相应足部抬升高于周围环境以避免碰撞。我们首先将此类无碰撞足集合定义为 $\mathcal{F}_{clear} = \{i : r_{f,i} > \max(H_{scan,i}), i \in I_{swing}\}$ ，其中 $H_{scan,i}$ 是第 i 个足部周围扫描高度的集合。则离地代价定义为

$$r_{fc} := \sum_{i \in I_{swing}} (\mathbb{1}_{\mathcal{F}_{clear}}(i) / |I_{swing}|) \in [0.0, 1.0]. \quad (15)$$

- 身体碰撞奖励 (r_{bc}): 我们希望惩罚机器人身体部位与地形之间的不良接触，以避免硬件损坏。

$$r_{bc} := -|I_{c,body} \setminus I_{c,foot}|. \quad (16)$$

- 目标平滑度奖励 (r_s): 对目标足部位置的二阶有限差分导数的幅值进行惩罚，以使生成的足部轨迹更加平滑。

$$r_s := -|(r_{f,d})_t - 2(r_{f,d})_{t-1} + (r_{f,d})_{t-2}|. \quad (17)$$

- 扭矩奖励 (r_τ): 我们对关节扭矩进行惩罚，以防止在部署过程中损坏关节执行器并降低能耗 ($\tau \propto$ 电流)。

$$r_\tau := -\sum_{i \in joints} |\tau_i|. \quad (18)$$

S5. 参数化地形

生成能够呈现代表性挑战（如足部打滑和足部卡陷）的训练环境至关重要。为了高效地合成随机地形，我们使用了程序化生成技术 [48]。该方法允许我们通过改变一组地形参数 $c_T \in \mathcal{C}$ 来生成大量不同的地形。下文将描述本工作中使用的三种地形生成器。地形可视化效果见图4B，参数空间 $\mathcal{C}$ 的定义见表S2。

- 丘陵地形基于Perlin噪声 [49]生成。该地形通过三个参数生成：粗糙度、Perlin噪声的频率以及Perlin噪声的振幅。输出高度图 hm 中每个元素的高度定义为 $hm[i, j] := \text{Perlin}(c_{T,2}, c_{T,3})[i, j] + U(-c_{T,1}, c_{T,1})$ 。在训练过程中，策略会在此地形上经历平滑斜坡和足部打滑。

- 台阶地形由高度随机的方形台阶组成。对于每个 $c_{T,1}$ 乘 $c_{T,1}$ 的区块，其高度从 $U(0, c_{T,2})$ 中采样。在训练过程中，策略会在此地形上经历离散的高度变化和足部卡陷。

- 楼梯地形是一种宽度和高度固定的阶梯。机器人初始位置位于楼梯中部的平坦段（见图4B）。

参数范围的设定考虑了机器人的运动学特性，例如台阶高度应低于腿长。在训练过程中，每个回合都会使用不同的随机种子重新生成地形。

S6. 自适应地形课程的定性评估

自适应课程的行为如图S1所示。图S1A-C聚焦于丘陵地形类型。该地形有三个参数：粗糙度、频率和振幅。可穿越性（公式3）与期望度（公式4）之间的关系如图S1A-B所示。不期望的地形要么过于简单，要么过于困难，如图S1A最左侧和最右侧面板所示。图S1B-C显示，粒子滤波器拟合了期望地形的潜在分布，该分布在频率-振幅边缘分布（中间图）中呈现弓形。图S1D聚焦于楼梯地形类型，展示了训练过程中地形参数的演化。粒子滤波器拒绝了代表短而陡峭台阶的参数（左上区域）。课程最初侧重于宽而浅的台阶（中间面板，特别是第50-60次迭代），随后将分布扩展至包含更窄的台阶（最右侧面板）。

S7. 不同情境下特权信息的重建

在图S2中，我们展示了不同情境下解码出的特权信息。图S2A显示了在穿越潮湿光滑的白板时，足部与地形之间的估计摩擦系数，如[电影S5](#)所示。当第一只足开始打滑时，估计值立即下降（i），在整个穿越过程中保持较低水平（ii），并在机器人返回正常地面约2秒后上升（iii）。外部扰动和地形信息也可以从TCN嵌入中重建。如图S2B所示，当施加未知的10kg负载时，解码器检测到向下的外力。在穿越如图S2C所示的茂密植被时，它检测到与运动方向相反的力，这促使策略进行反作用并推开植被。在图S2C和图S2D所示的自然地形中，高程估计的不确定性显著较高，这表明TCN策略编码了地形的粗糙度。

S8. 循环神经网络学生策略

我们使用TCN架构作为本体感受策略 [22]。为了进行比较，我们还评估了带有门控循环单元（GRU）的循环网络 [50]。架构在表格S5和S6中详细说明。训练GRU学生策略的损失函数定义为

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t))^2 + (\bar{l}_t(o_t, x_t) - l_t(o_t))^2. \quad (19)$$

为了提高训练的性能和计算效率，我们实现了截断时间反向传播（截断BPTT）[51]。

图S3展示了在图5A所示诊断设置上的性能表现。总体而言，基于GRU的控制器的性能介于TCN-20与TCN-100之间。在斜坡设置中，其性能与TCN-100相当，但在台阶实验中，基于GRU的控制器未能达到TCN-100的性能水平。

The chief advantage of the TCN is in training efficiency. The training time for the TCN is much faster in comparison to the GRU. The computation times are reported in Table [S1](#).

S9. Ablation of the latent representation loss for student training

We examine the effect of the second term in the loss function for student policy training presented in Eq. 1, which is a squared error loss for the latent vector l_t . As a baseline, we train a student policy using the following loss function:

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t, H))^2, \tag{20}$$

which simply imitates the output of the teacher.

The result is reported in Fig. S3 as ‘TCN-100 naive IL’. The performance is comparable in the uniform slope setting and under external disturbances. On the other hand, the ablated version has lower success rates on steps.

TCN的主要优势在于训练效率。与GRU相比，TCN的训练时间要快得多。计算时间报告于表格[S1](#)中。

S9. 学生策略训练中潜在表示损失的消融实验

我们检验了公式1中学生策略训练损失函数中第二项的影响，该项是潜在向量 l_t 的平方误差损失。作为基线，我们使用以下损失函数训练学生策略：

$$\mathcal{L} := (\bar{a}_t(o_t, x_t) - a_t(o_t, H))^2, \tag{20}$$

该函数仅模仿教师的输出。

结果如图S3中‘TCN-100朴素模仿学习’所示。在均匀斜坡设置和外部干扰下，性能相当。另一方面，消融版本在台阶上的成功率较低。

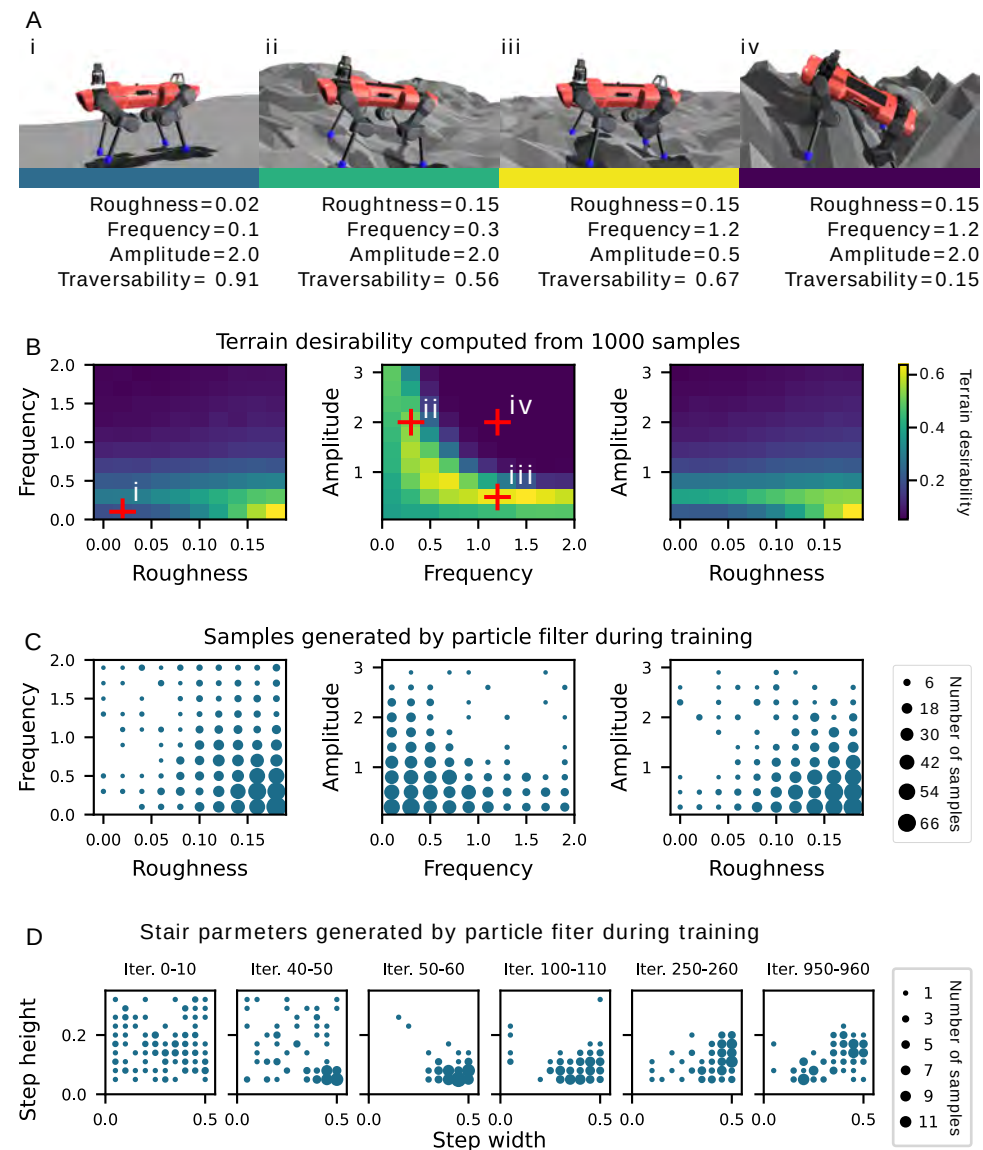
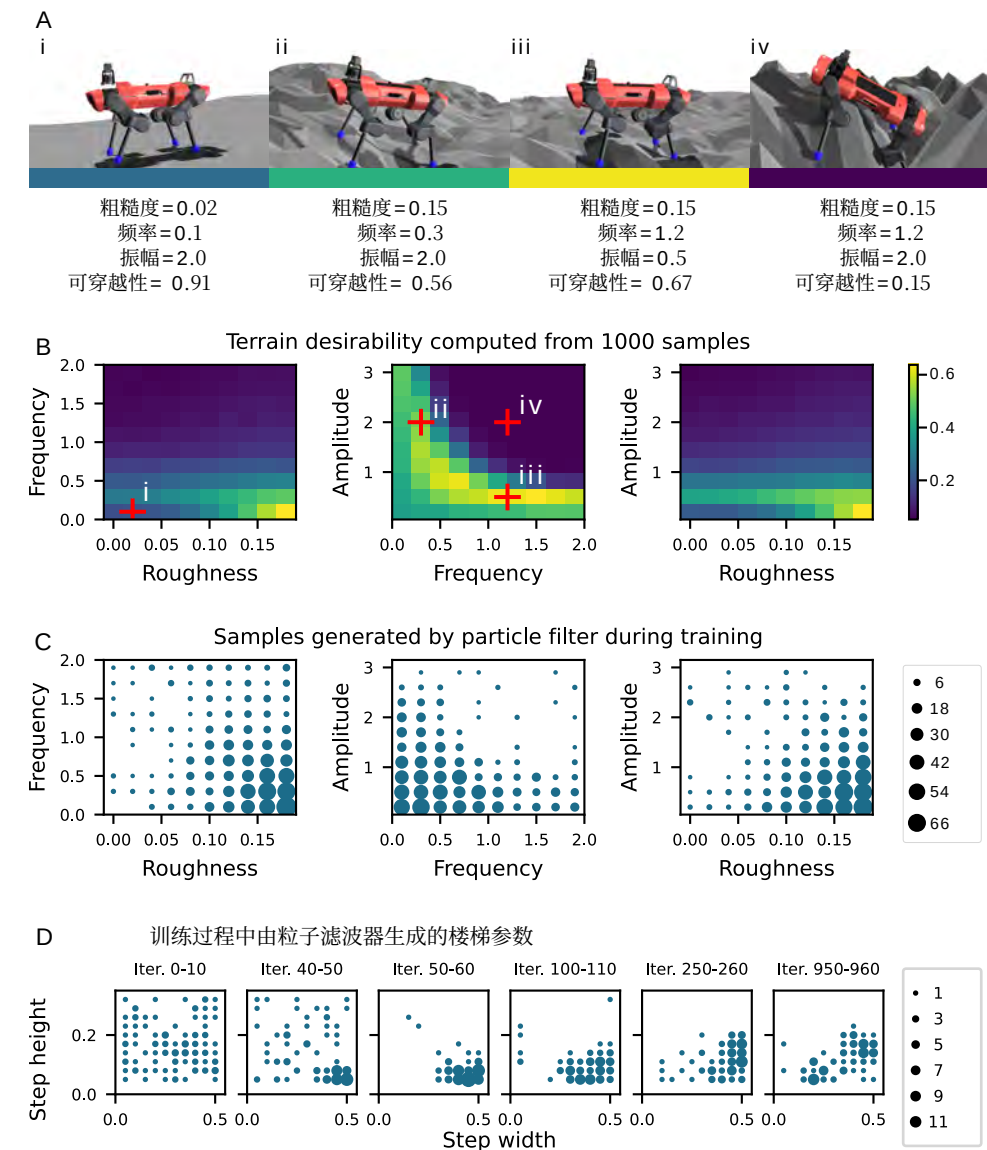


Fig. S1. Illustration of the adaptive curriculum. (A) Examples of *Hills* terrains. The color bar indicates desirability; dark blue represents low desirability. (B) Terrain desirability estimated from 1000 trajectories generated by a fully trained teacher policy. The red crosses correspond to the examples presented in A. (C) The distribution of terrain profiles sampled by the particle filter during the last 100 iterations of teacher training. (D) Evolution of *Stairs* terrain parameters during training.



图S1. 自适应课程示意图。 (A) 丘陵地形的示例。色标表示期望度；深蓝色代表低期望度。(B) 根据完全训练好的教师策略生成的1000条轨迹估算的地形期望度。红色十字标记对应A中展示的示例。(C) 在教师训练最后100次迭代中由粒子滤波器采样的地形剖面分布。(D) 训练过程中楼梯地形参数的演化。

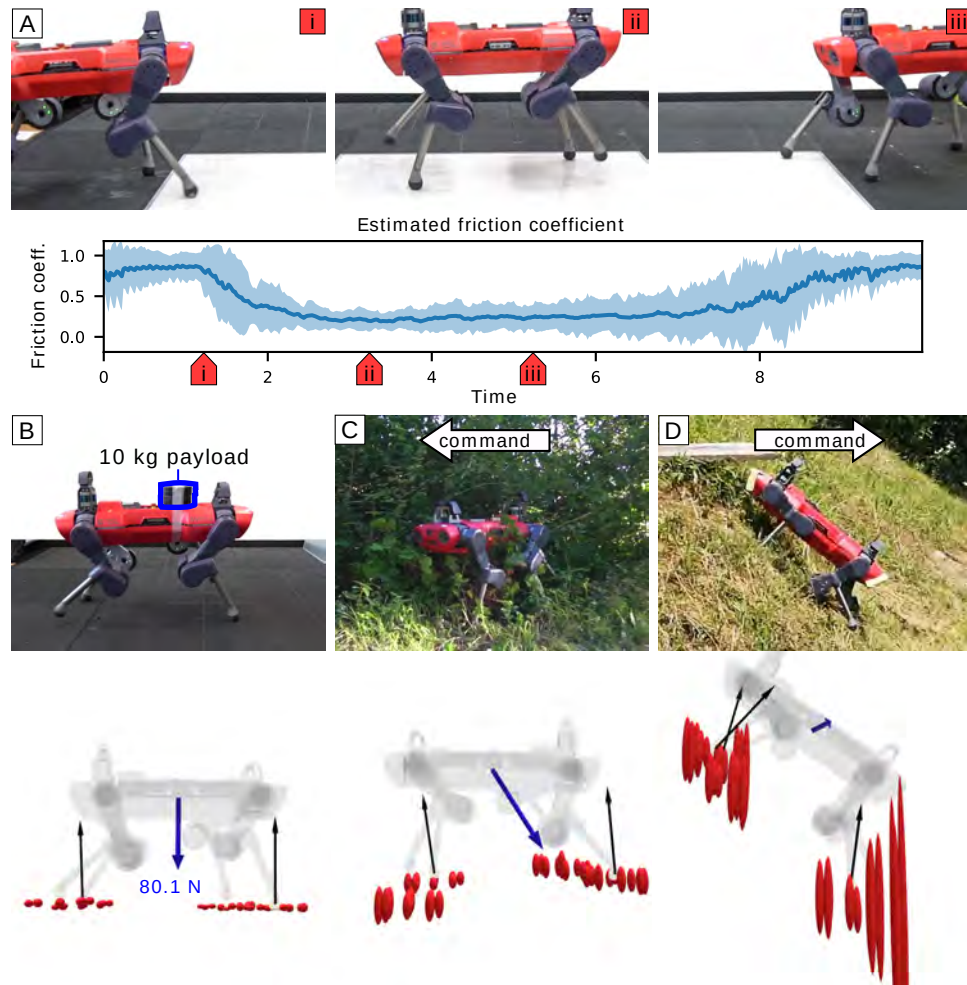


Fig. S2. Reconstructed privileged information in different situations. (A) Estimated friction coefficient between the feet and the terrain while traversing a wet whiteboard. The shaded area denotes 95 % confidence interval. (B-D) Reconstruction of the external disturbance and terrain information in different scenarios. Blue arrow: estimated external force applied to the torso. Red ellipsoid: estimated terrain shape around the foot. The center of the ellipsoid refers to the estimated terrain elevation and the vertical length represents uncertainty (1 standard deviation). For each foot, 8 ellipsoids are symmetrically placed along a circle with 10 cm radius. Black arrow: terrain normal at the in-contact foot.

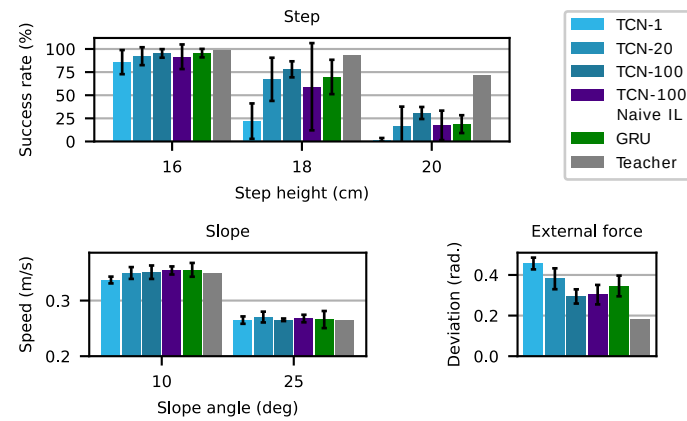
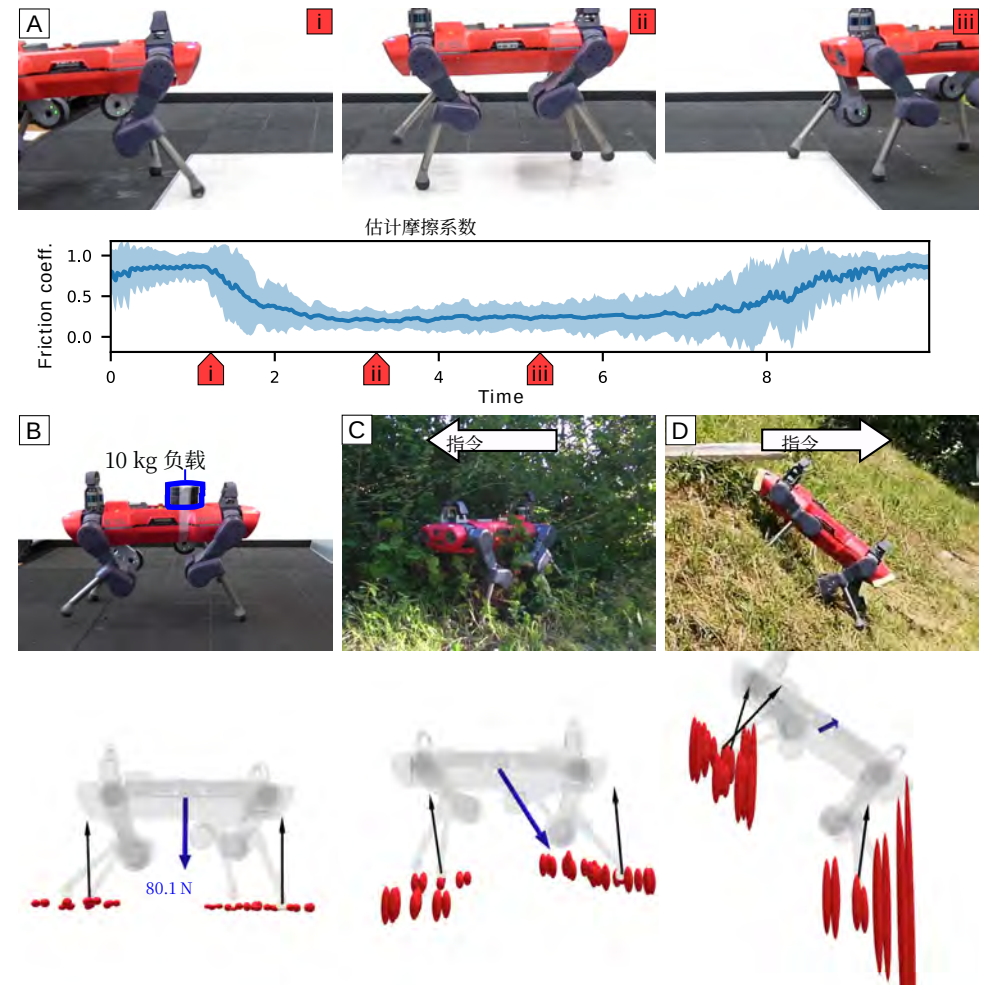
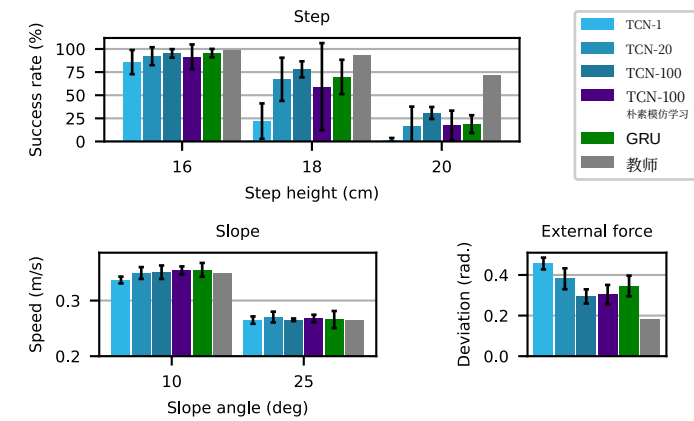


Fig. S3. Comparison of neural network architectures for the proprioceptive controller. We trained each model 5 times using different random seeds. The error bars denote 95% confidence intervals. 'TCN-100 naive IL' denotes the TCN-100 network trained using a naive imitation learning method without the latent representation loss (Eq. 19).



图S2. 不同情境下重建的特权信息。 (A) 穿越湿白板时足部与地形之间的估计摩擦系数。阴影区域表示95%置信区间。(B-D) 不同场景下外部扰动和地形信息的重建。蓝色箭头：施加于躯干的估计外力。红色椭球体：足部周围估计的地形形状。椭球体中心表示估计的地形高程，垂直长度代表不确定性（1个标准差）。每个足部有8个椭球体沿半径为10厘米的圆对称放置。黑色箭头：接触足处的地形法线。



图S3. 本体感受控制器的神经网络架构比较。 我们使用不同的随机种子对每个模型进行了5次训练。误差条表示95%置信区间。'TCN-100朴素模仿学习'表示使用朴素模仿学习方法训练的TCN-100网络，该方法不包含潜在表示损失（公式19）。

Name	Time
Teacher policy training	≈ 12 hrs
Student policy training	≈ 4 hrs
Adaptive terrain curriculum	2.9 s

Table S1. Computation time for training. The TCN-100 architecture is used for the student policy. The training is conducted on a desktop machine with i7-8700K CPU and a Geforce RTX 2080 GPU.

Terrain	grid size	friction coefficient	parameters (c_T)	range
<i>Hills</i>	0.2 m	$\mathcal{N}(0.7, 0.2)$	roughness (m)	[0.0, 0.05]
			frequency	[0.2, 1.0]
			amplitude (m)	[0.2, 3.0]
<i>Slippery Hills</i>	0.2 m	$\mathcal{N}(0.3, 0.1)$	roughness (m)	[0.0, 0.05]
			frequency	[0.2, 1.0]
			amplitude (m)	[0.2, 3.0]
<i>Steps</i>	0.02 m	$\mathcal{N}(0.7, 0.2)$	step width (m)	[0.1, 0.5]
			step height (m)	[0.05, 0.3]
<i>Stairs</i>	0.02 m	$\mathcal{N}(0.7, 0.2)$	step width (m)	[0.1, 0.5]
			step height (m)	[0.02, 0.2]

Table S2. Parameter spaces $\mathcal{C}$ for simulated terrains. $\mathcal{N}(m, d)$ denotes that the value is sampled from the Gaussian distribution of mean m and standard deviation d . The friction coefficient is clipped to be above 0.1.

Parameter	value
Number of particles ($N_{particle}$)	10 per terrain type
Transition probability ($p_{transition}$)	0.8
Trajectories per particle (N_{traj})	6
Update rate of the terrain parameters ($N_{evaluate}$)	10
Probability of sampling from replay memory (P_{replay})	0.05

Table S3. Hyperparameters for automatic terrain curriculum.

Name	Time
教师策略训练	≈ 12 小时
学生策略训练	≈ 4 小时
自适应地形课程	2.9 秒

表S1. 训练的计算时间。学生策略采用TCN-100架构。训练在一台配备i7-8700K CPU和一台配备i7-8700K CPU和Geforce RTX 2080 GPU的台式机上进行。

地形	网格尺寸	摩擦系数	参数 (c_T)	范围
丘陵	0.2 米	$\mathcal{N}(0.7, 0.2)$	粗糙度 (米)	[0.0, 0.05]
			频率	[0.2, 1.0]
			振幅 (m)	[0.2, 3.0]
Slippery Hills	0.2 m	$\mathcal{N}(0.3, 0.1)$	粗糙度 (m)	[0.0, 0.05]
			频率	[0.2, 1.0]
			振幅 (米)	[0.2, 3.0]
台阶	0.02 米	$\mathcal{N}(0.7, 0.2)$	步宽 (米)	[0.1, 0.5]
			台阶高度 (米)	[0.05, 0.3]
楼梯	0.02 米	$\mathcal{N}(0.7, 0.2)$	步宽 (米)	[0.1, 0.5]
			台阶高度 (米)	[0.02, 0.2]

表S2. 参数空间 $\mathcal{C}$ 用于模拟地形。 $\mathcal{N}(m, d)$ 表示该值从均值为 m 、标准差为 d 的高斯分布中采样。摩擦系数被限制在0.1以上。

参数	值
粒子数量 ($N_{particle}$)	每种地形类型10个
转移概率 ($p_{transition}$)	0.8
每条粒子的轨迹数 (N_{traj})	6
地形参数更新率 ($N_{evaluate}$)	10
从回放记忆采样的概率 (P_{replay})	0.05

表S3. 自动地形课程的超参数。

Data	dimension	x_t	o_t	h_t
Desired direction ($((^B_{IB}\hat{v}_d)_{xy})$	2		✓	✓
Desired turning direction ($((^B_{IB}\hat{\omega}_d)_z)$	1		✓	✓
Gravity vector (e_g)	3		✓	✓
Base angular velocity ($^B_{IB}\omega$)	3		✓	✓
Base linear velocity ($^B_{IB}v$)	3		✓	✓
Joint position/velocity ($\theta_i, \dot{\theta}_i$)	24		✓	✓
FTG phases ($\sin(\phi_i), \cos(\phi_i)$)	8		✓	✓
FTG frequencies ($\dot{\phi}_i$)	4		✓	✓
Base frequency (f_o)	1		✓	
Joint position error history	24		✓	
Joint velocity history	24		✓	
Foot target history ($(r_{f,d})_{t-1,t-2}$)	24		✓	
Terrain normal at each foot	12	✓		
Height scan around each foot	36	✓		
Foot contact forces	4	✓		
Foot contact states	4	✓		
Thigh contact states	4	✓		
Shank contact states	4	✓		
Foot-ground friction coefficients	4	✓		
External force applied to the base	3	✓		

Table S4. State representation for proprioceptive controller (top) and the privileged information (bottom).

Layer	Teacher		TCN-N Student		GRU Student		Decoder
input	o_t	x_t	o_t	h (60×N)	o_t	o_t	$\langle o_t, l_t \rangle$
1	id	tanh(72)	id	1D conv dilation 1	id	GRU(68)	relu(196)
2	id	tanh(64)	id	1D conv stride 2	concatenate		Output
3	concatenate		id	1D conv dilation 2	tanh(256)*		-
4	tanh(256)*		id	1D conv stride 2	tanh(128)*		-
5	tanh(128)*		id	1D conv dilation 4	tanh(64)*		-
6	tanh(64)*		id	1D conv stride 2	Output*		-
7	Output*		id	tanh(64)	-		-
8	-		concatenate		-		-
9	-		tanh(256)*		-		-
10	-		tanh(128)*		-		-
11	-		tanh(64)*		-		-
12	-		Output*		-		-

Table S5. Neural network architectures. Unless specified otherwise, the dilation and stride are 1 for convolutional layers. The filter size is fixed to 5. The layers marked with * are copied from the teacher to learners after the teacher training. id refers to the identity map. The TCN-N architecture uses dilated causal convolution [22]. Each convolutional layer is followed by a relu activation function.

Data	维度	x_t	o_t	h_t
期望方向 ($((^B_{IB}\hat{v}_d)^{xy})$	2		✓	✓
期望转向方向 ($((^B_{IB}\hat{\omega}_d)_z)$	1		✓	✓
重力向量 (e_g)	3		✓	✓
基座角速度 ($^B_{IB}\omega$)	3		✓	✓
基座线速度 ($^B_{IB}v$)	3		✓	✓
关节位置/速度 ($\theta_i, \dot{\theta}_i$)	24		✓	✓
FTG相位 ($\sin(\phi_i), \cos(\phi_i)$)	8		✓	✓
FTG频率 ($\dot{\phi}_i$)	4		✓	✓
基频 (f_o)	1		✓	
关节位置误差历史	24		✓	
关节速度历史	24		✓	
足部目标历史 ($(r_{f,d})^{t-1,t-2}$)	24		✓	
各足地形法线	12	✓		
各足周围高度扫描	36	✓		
足部接触力	4	✓		
足部接触状态	4	✓		
大腿接触状态	4	✓		
小腿接触状态	4	✓		
足地摩擦系数	4	✓		
施加于基座的外力	3	✓		

表格 S4. 本体感受控制器的状态表示（上）与特权信息（下）。

层	教师		TCN-N学生		GRU学生		解码器
输入	o_t	x_t	o_t	h (60×N)	o_t	o_t	$\langle o_t, l_t \rangle$
1	id	双曲正切函数(72)	id	1D卷积 膨胀 1	id	GRU(68)	relu(196)
2	id	双曲正切函数(64)	id	一维卷积步长2	拼接		输出
3	拼接		id	一维卷积膨胀2	双曲正切函数(256)*		-
4	双曲正切函数(256)*		id	一维卷积步长2	双曲正切函数(128)*		-
5	双曲正切函数(128)*		id	一维卷积膨胀4	双曲正切函数(64)*		-
6	双曲正切函数(64)*		id	一维卷积步长2	输出*		-
7	输出*		id	双曲正切函数(64)	-		-
8	-		拼接		-		-
9	-		双曲正切函数(256)*		-		-
10	-		双曲正切函数(128)*		-		-
11	-		双曲正切函数(64)*		-		-
12	-		输出*		-		-

表S5. 神经网络架构。除非另有说明，卷积层的膨胀和步长均为1。滤波器大小固定为5。标记有*的层在教师训练后从教师复制到学生网络。恒等映射指的是恒等映射。TCN-N架构使用膨胀因果卷积 [22]。每个卷积层后都接一个ReLU激活函数。

Model	seq. length	# channels	# param.	SGD time (s)
TCN-1	1	60	161960	9.22e-3 (±1.78e-3)
TCN-20	20	44	158300	2.11e-2 (±1.24e-3)
TCN-100	100	34	158070	5.07e-2 (±1.94e-3)
GRU	100*	-	159640	1.52e-1 (±1.89e-2)

Table S6. Network parameter settings and the training time for student policies. SGD time refers to the computation time required for one stochastic gradient descent update with the batch size given in Table S8. The computation times are presented as empirical means with standard deviations. *The sequence length for the GRU network stands for the sequence length used for Truncated BPTT [51].

Parameter	Value
discount factor	0.995
KL-d threshold	0.01
max. episode length	400
CG damping	1e-1
CG iteration	50
discount factor	0.995
batch size	80000
total iterations	10000

Table S7. Hyperparameters for teacher policy training.

Parameter	TCN-N	GRU
initial learning rate	5e-4	2e-4
learning rate decay	exp(0.995, 100)	
max. episode length	400	
batch size	20000	10000
minibatches	5	
epochs	4	
total iteration	4000	

Table S8. Hyperparameters for student policy training. exp(a,b) denotes exponential decay, which is defined as $lr_0 * a^{updates/b}$. The Adam [52] optimizer is used.

Parameter	values
initial learning rate	1e-4
learning rate decay	exp(0.99, 100)
batch size	20000
minibatches	2
epochs	10
total iteration	1000
weight decay	l_2 -norm, 1e-4

Table S9. Hyperparameters for decoder training. exp(a,b) denotes exponential decay, which is defined as $lr_0 * a^{updates/b}$. The Adam [52] optimizer is used.

模型	序列长度	通道数	参数量	SGD时间（秒）
TCN-1	1	60	161960	9.22e-3 (±1.78e-3)
TCN-20	20	44	158300	2.11e-2 (±1.24e-3)
TCN-100	100	34	158070	5.07e-2 (±1.94e-3)
GRU	100*	-	159640	1.52e-1 (±1.89e-2)

表S6. 学生策略的网络参数设置与训练时间。 SGD时间指在表S8给定的批量大小下，执行一次随机梯度下降更新所需的计算时间。计算时间以经验均值±标准差的形式呈现。*GRU网络的序列长度指用于截断BPTT的序列长度 [51]。

参数	数值
折扣因子	0.995
KL散度阈值	0.01
最大回合长度	400
共轭梯度阻尼	1e-1
共轭梯度迭代	50
折扣因子	0.995
批量大小	80000
总迭代次数	10000

表S7. 教师策略训练的超参数。

参数	TCN-N	GRU
初始学习率	5e-4	2e-4
学习率衰减	exp(0.995, 100)	
最大回合长度	400	
批量大小	20000	10000
小批次	5	
轮次	4	
总迭代次数	4000	

表S8. 学生策略训练的超参数。 exp(a,b) 表示指数衰减，其定义为 $lr_0 * a^{updates/b}$ 。使用 Adam [52] 优化器。

参数	数值
初始学习率	1e-4
学习率衰减	exp(0.99, 100)
批量大小	20000
小批次	2
轮次	10
总迭代次数	1000
权重衰减	l_2 -范数, 1e-4

表S9. 解码器训练的超参数。 exp(a,b)表示指数衰减，其定义为学习率 $_0 * a^{updates/b}$ 。Adam [52] 优化器被使用。